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Cryptocurrencies as shock transmitters: dynamic connectedness, hedging strategies, and portfolio management across financial markets for higher-order moments

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Abstract

This study explores the higher-order moments of connectedness among cryptocurrency, commodity, bond, and stock markets from April 19, 2017, to December 29, 2023, on the basis of the GARCH-SK and TVP-VAR models. The findings reveal that Bitcoin and Ethereum act as significant net shock transmitters, especially during major events such as the COVID-19 pandemic and the Russia–Ukraine conflict. After mid-2021, these cryptocurrencies transitioned from net receivers to net transmitters of volatility owing to rising economic and geopolitical risks. These insights assist in portfolio diversification strategies. By combining shock transmitters with shock-resilient cryptocurrencies, investors can enhance their risk profiles. Diversification opportunities shift during financial crises, making it crucial to focus on shock transmitters, which are less influenced by various risk factors. Additionally, the study highlights cryptocurrencies as potential safe havens compared with traditional assets such as gold, bonds, and stocks, which often maintain or appreciate value during market stress. TVP-VAR-informed dynamic portfolio reallocation can improve risk-adjusted returns and lower volatility, aiding in capital preservation during high TCI periods. Overall, our findings suggest that portfolios that include cryptocurrencies generally outperform those that do not, emphasizing their role as effective diversifiers in portfolio optimization and financial stability.

Keywords: Risk connectedness, TVP-VAR connectedness, Higher-order moments, Financial markets

JEL Classification: G14, G15, F3

Introduction

Cryptocurrencies have ushered in a new era within the financial landscape, challenging traditional paradigms and necessitating a reevaluation of market dynamics. As digital assets continue to gain prominence, understanding their impact on conventional financial markets has become increasingly critical. During periods of economic uncertainty, investors typically seek refuge in stable, safe-haven assets to mitigate the risks associated with market downturns. These assets provide a sense of security and are essential for safeguarding investments. However, cryptocurrencies, operating independently of traditional financial systems and monetary policies, have the potential to serve as safe-haven assets during times of market turmoil and global uncertainty (Ji et al. 2020; Shahzad et al. 2019, 2020; Mariana et al. 2021; Ustaoglu 2022; Yousaf and Yarovaya 2022; Echaust et al. 2024; Hsu et al. 2024). Research suggests that cryptocurrencies can be effective tools for portfolio diversification, offering a hedge against sovereign risk (Corbet et al. 2018; Kajtazi and Moro 2019; Zeng et al. 2020; Bakry et al. 2021; Joshi et al. 2022). Additionally, cryptocurrencies can function as hedging assets, reducing losses or increasing profits (Demir et al. 2018; Guesmi 2011; Naeem et al. 2020). Nevertheless, the inherent volatility of cryptocurrency prices introduces significant risks, and indiscriminately adding cryptocurrencies to investment portfolios during financial crises may amplify systemic risk across the financial system (Conlon et al. 2020; Goodell and Goutte 2021; Ji et al. 2020; Grobys 2021). Moreover, the rapid growth of decentralized finance (DeFi) during the pandemic has sparked debates among policymakers regarding the efficacy of cryptocurrencies as a hedge against risk. The literature explores the relationships between cryptocurrencies and global stock markets (Nguyen 2022; Zhao and Zhang 2023), commodity markets (Bouri et al. 2018b; Rehman and Apergis 2019; Rehman and Vo 2020; Mo et al. 2022; Echaust et al. 2024), and bond markets (Bouri et al. 2018a). In this context, it is crucial for investors and policymakers to thoroughly investigate the interconnectedness between cryptocurrencies and other markets.

The economic uncertainties triggered by the pandemic and their effects on financial markets have led investors to seek alternative assets for hedging against risk. Cryptocurrencies, particularly Bitcoin, have gained attention for their potential to offer hedging benefits against stock market declines (Baur et al. 2018; Al-Yahyaee et al. 2019) and for their independence from traditional assets (Baur et al. 2018; Al-Yahyaee et al. 2019) and monetary policies (Narayan et al. 2019), which can help reduce portfolio risks during financial market turmoil (Gil-Alana et al. 2020; Al-Yahyaee et al. 2019). However, the high volatility of Bitcoin poses challenges for risk management, raising questions about its suitability as a safe-haven asset. Despite these divergent views, current research has yet to provide definitive evidence on whether cryptocurrencies can reliably serve as safe-haven assets or if they function merely as diversification tools (Yermack 2024; Shahzad et al. 2019). This paper examines the interconnectedness between cryptocurrencies and traditional financial markets (i.e., commodity, bond, and stock markets), focusing on the implications for portfolio management and market stability.

Cryptocurrencies possess unique characteristics that distinguish them from traditional assets such as stocks, bonds, and commodities. They can function as a medium of exchange without the influence of central banks' stabilization policies, making them ideal for portfolio diversification. The returns of cryptocurrencies are typically uncorrelated with other asset classes, further enhancing their diversification benefits. Three main channels drive the inclusion of cryptocurrencies in financial markets: the monetary system, behavioral theories, and portfolio diversification (Schilling and Uhlig 2019; Maitra et al. 2022). With respect to the monetary system, cryptocurrencies significantly affect an economy through three channels: inflation, monetary aggregates, and exchange rates. The adoption of cryptocurrencies as a replacement for traditional currencies profoundly changes the function of money and slows its circulation. This shift challenges traditional monetary theories, potentially requiring updates or replacements to keep pace with the evolving financial landscape (Narayan et al. 2019). Behavioral channels can elucidate the differences between cryptocurrencies and traditional financial markets, as the gradual information diffusion hypothesis and investor conservatism can influence price movements across assets. Narayan et al. (2019) propose that cryptocurrencies may behave differently from traditional stocks, with the information diffusion hypothesis (Hong and Stein 1999) offering insights into this disparity. Furthermore, investors in traditional markets often exhibit biases that affect their reactions to market shocks, leading to under- or overreactions to changes in other assets. This behavior is more prevalent in traditional asset markets and may not apply to cryptocurrencies, which may behave differently (Narayan and Sharma 2011). This variability among cryptocurrency investors makes cryptocurrencies appealing assets for portfolio diversification. Their characteristics differ from those of traditional assets such as stocks and bonds, resulting in a diverse business cycle, which makes them attractive to investors seeking to diversify their portfolios. Additionally, the variables influencing cryptocurrency returns differ from those affecting traditional assets, providing further diversification benefits. Consequently, cryptocurrencies can serve as valuable additions to investment portfolios (Kang et al. 2019).

In light of the above information, this study examines the dynamic connectedness between cryptocurrencies and traditional financial markets via a time-varying parameter vector autoregression (TVP-VAR) model. By analyzing daily price and volume data from cryptocurrency and financial markets between April 19, 2017, and December 29, 2023, we aim to capture the evolving nature of their interplay. Cryptocurrencies, particularly Bitcoin, Ethereum, and Tether, have experienced rapid growth and volatility, drawing the attention of investors and policymakers. Their decentralized nature and technological foundations present both opportunities and challenges, raising questions about their role within the broader financial ecosystem. Traditional financial markets, represented in this study by the DAX, S&P 500, and SENSEX indices, have also been influenced by the rise of digital assets, necessitating a deeper analysis of their relationship. This study provides insights into volatility transmission and the roles of different markets as information transmitters or receivers. Furthermore, it explores practical applications for investors, proposing hedging strategies and minimum connectedness portfolio weights to help navigate the complexities of a market landscape increasingly influenced by cryptocurrencies.

This study comprehensively analyzes the returns and volatility connectedness between cryptocurrencies and traditional financial markets, as well as their hedging and safe haven behaviors. In doing so, we aim to shed further light on the potential impact of cryptocurrencies on the broader financial system. To contribute to the literature, we pose the following research questions:

- How much would adding cryptocurrencies to a portfolio of traditional assets, such as commodities, bonds, and stocks, enhance hedging efficiency?
- Can investors consider cryptocurrencies safe havens against extreme stock market shocks?

- When these assets are most effective in reducing stock market risk, considering the time-varying nature of their hedging effectiveness?
- Do cryptocurrencies, gold, and oil warrant attention as hedging instruments, given their widespread use by investors?

Our findings reveal a moderate level of connectedness, with significant implications for portfolio diversification and risk management. The total connectedness index (TCI) indicates that interdependence among the markets is relatively low, offering potential benefits for investors seeking to hedge against market-specific risks. The analysis of major events, including the COVID-19 outbreak, the Russia–Ukraine conflict, and the Silicon Valley Bank crisis, underscores the importance of understanding dynamic connectedness during periods of market stress. Moreover, our portfolio analysis indicates the existence of a dynamic network that presents opportunities for diversification. Our research demonstrates that incorporating Bitcoin and Ethereum into a portfolio can mitigate the volatility of other assets within the portfolio. These findings highlight the relevance of cryptocurrencies as effective tools for diversifying stock market investments. Consequently, a prudent portfolio manager should consider the hedging effectiveness and connectedness of cryptocurrencies when making investment decisions.

This study contributes significantly to the literature. First, it enhances the growing body of research on the impact of cryptocurrency adoption on traditional financial markets. The role of cryptocurrencies in diversifying equity market risk during the pandemic remains a debated topic due to conflicting evidence. Existing studies (Basher and Sadorsky 2016; Antonakakis et al. 2018; Maitra et al. 2022; Nham 2022; Harb et al. 2024; Echaust et al. 2024) examine the connectedness between cryptocurrencies, commodities, and stock markets to develop effective portfolio allocations. However, we also consider spillovers between cryptocurrencies and bond markets to highlight their role in hedging against monetary policy shocks. Second, we investigate the connectedness of returns and volatility between cryptocurrencies and traditional financial markets to assess market asymmetry during times of economic and geopolitical risk. This dynamic approach requires investors to adapt their portfolio strategies to evolving market conditions, emphasizing the importance of efficient investment decision-making for portfolio diversification and risk management. Additionally, monitoring time-varying higher-order moments can serve as a crucial market indicator, capturing changing risk patterns and enabling policymakers to respond promptly to market instability. Furthermore, we seek to validate and explore the impact of economic and geopolitical risks, such as the COVID-19 outbreak, the Russia–Ukraine war, and the Silicon Valley Bank crisis, on the spillover effects of higher-order moments. While previous studies (Ji et al. 2019, 2021; Bouri et al. 2021a; Kumar et al. 2022) have focused on return and volatility spillovers among major cryptocurrencies, our study aims to provide a comprehensive analysis of the interconnectedness of returns and volatility among cryptocurrencies, crude oil, gold, bonds, and stocks. We also intend to expand the evaluation of hedging strategies beyond risk reduction to consider their profitability for investors, addressing the current literature's limited focus on assessing hedging strategies solely on the basis of their effectiveness in reducing risk.

Third, this study provides comprehensive empirical assessments of higher-moment spillovers between cryptocurrencies and traditional asset classes. By examining time-varying skewness and kurtosis dynamics, we uncover how asymmetries and tail risks are transmitted across markets, particularly during periods of financial stress. These findings advance the literature on portfolio risk management by demonstrating the importance of incorporating higher moments in hedging and diversification strategies.

Finally, we employ a method developed by Antonakakis et al. (2020) known as time-varying parameter vector autoregression (TVP-VAR) with stochastic volatility, which is based on Diebold and Yilmaz's (2012, 2014) framework, to investigate the dynamic connectedness measures of both cryptocurrency and traditional financial markets. The reason we use this model is that the TVP-VAR method offers several key advantages over the traditional VAR model. It allows for connectedness analysis across different time horizons, enabling the examination of dynamics between series in both the short and long run. This method effectively incorporates timing into the calculation of connectedness levels, addressing the effects of economic shocks. Additionally, it captures the uncertainty of the modeled series by reporting interconnectedness across various periods, which aids in identifying potential nonlinear relationships. The TVP-VAR method also facilitates h-step-ahead prediction error variance decomposition, providing significant benefits: (i) it calculates total and net total directional connectedness, delivering a precise assessment of market uncertainty; (ii) it highlights potential asymmetries among series through time-varying results; and (iii) it reduces biases by accounting for the impacts of economic shocks (Chishti et al. 2024a,b; Zhu et al. 2024; Patel et al. 2024; Xia et al. 2024). TVP-VAR-based dependence analysis can help reduce investors' risk exposure during market turbulence and reveal the characteristics of specific cryptocurrencies that can act as hedges against portfolio risk within the stock market. Understanding the transmission of return and volatility risks during both upward and downward market movements can influence investors' decisions regarding risk-on and risk-off strategies. This model provides a flexible and robust approach to capturing potential changes in the underlying data structure. This eliminates the need for a fixed rolling window size or the sacrifice of observations in the calculation of dynamic connectivity measures, as it does not involve rolling window analysis. Additionally, it is robust to the presence of outliers. Through this approach, we calculate net pairwise connectedness, which helps identify spillover mechanisms among the cryptocurrency, commodity, bond, and stock markets. This allows for a clearer understanding of each market's role within the system as either a volatility sender or receiver. This is particularly valuable in times of economic uncertainty, where understanding the dynamics of different markets can enable investors to make more informed decisions (Pham and Do 2022; Lu et al. 2023). We compare the findings of the TVP-VAR with stochastic volatility to those obtained from various rolling window (150, 200, and 250) and different prior (uninformative, Minnesota, and Bayesian) settings. These findings have the potential to reshape our understanding of financial markets. We also evaluate hedging strategies on the basis of minimum connectedness and minimum variance portfolios, which go beyond simply measuring their effectiveness in reducing risk relative to single-asset portfolios. These strategies can indeed provide insights into how cryptocurrencies respond to external shocks. Diversifying investments across different cryptocurrencies and other assets may be beneficial for managing risk and maximizing returns. By monitoring the performance of various assets within a portfolio, investors can make informed decisions about when to buy, sell, or hold. Additionally, our paper contributes to the literature by examining spillovers across cryptocurrency and traditional financial markets at higher realized moments, moving beyond the typical focus on returns and volatility. Recent studies indicate that higher realized moments can provide valuable insights, especially when returns are not normally distributed (Bonato et al. 2020; Bouri et al. 2021a, b, c, d; Zhang et al. 2023; Hao and Pham 2024). We highlight the importance of skewness and kurtosis spillovers, which capture return asymmetry and extreme events. These higher-order moment spillovers can influence market stability and should be studied in detail. Moreover, including time-varying skewness in spillover analyses is crucial for investor decisions, as it relates to the asymmetry in return distributions and extreme risks. Investors generally prefer positively skewed portfolios (Kraus and Litzenberger 1976), and understanding skewness spillovers helps clarify market interconnections and informational efficiency regarding cross-border risk absorption. Del Brio et al. (2017) emphasize the need to analyze higher moments, such as skewness and kurtosis, to understand globalization effects in financial markets. These measures capture asymmetric return behavior and the transmission of extreme events, offering deeper insights into systemic risk and portfolio vulnerability. This study addresses this gap by employing the GRACHSK and TVP-VAR methods to explore the asymmetric and tail risk connectedness between cryptocurrency, stock, bond, crude oil, and gold markets.

Literature

The inherent volatility and interconnectedness of the cryptocurrency market have spurred significant academic interest in understanding its dynamics and implications for traditional financial markets. The literature predominantly addresses strategies for portfolio management and hedging within cryptocurrencies, the role of uncertainties in market dynamics, and the interconnectedness of volatility and returns between digital and traditional assets. This review aims to provide a comprehensive overview of current research in the field, highlighting the complexities and opportunities presented by integrating cryptocurrencies into the broader financial landscape.

Existing studies have investigated the interrelations between major cryptocurrencies (Yi et al. 2018; Bouri et al. 2021b,c; Polat 2023; Poddar et al. 2023) and their impact on market dynamics (Kurka 2019; Charfeddine et al. 2020; Chemkha et al. 2021), which is crucial for understanding the risk factors and potential contagion effects characterizing the cryptocurrency ecosystem. Yi et al. (2018) analyze volatility spillovers among cryptocurrencies, identifying "mega-cap" cryptocurrencies such as Bitcoin as significant propagators of volatility. Bouri et al. (2021b,c) introduce a quantile-based method to assess market connectedness, noting an increased transmission of return shocks under extreme market conditions. Aslanidis et al. (2021) explore dynamic market linkages among cryptocurrencies and find a significant increase in linkages for both returns and volatilities, thus offering insights into the evolving interconnectedness of cryptocurrencies. Mensi et al. (2021) investigate the frequency connectedness of volatility differences among eight prominent cryptocurrencies, assessing diversification benefits and downside risk reductions via the Diebold and Yilmaz (2012, 2014) and Barunik and Křehlík

(2018) connectedness approaches. Their study reveals intensified dynamic spillovers post-2017, with Bitcoin, Ethereum, and Litecoin identified as net transmitters of risk, and highlights that short-term risk spillovers were more pronounced than medium- and long-term spillovers were. Polat (2023) examines dynamic volatility connectedness among major cryptocurrencies via TVP-VAR approaches, identifying Ethereum, Bitcoin, and Link as primary propagators and recipients of shocks, with the most intense volatility interdependencies occurring in the short term.

The integration of cryptocurrency markets with traditional financial systems and the resulting spillover effects are central themes in the literature. These studies focus on quantifying market integration and analyzing the returns and volatility spillover mechanisms between different asset classes. Understanding these dynamics is essential for navigating the rapidly evolving financial landscape. The economic uncertainties triggered by the pandemic and the resulting financial market shocks have led investors to seek alternative assets for hedging. Cryptocurrencies have gained popularity, prompting researchers to explore their potential as safe havens, hedges, or diversifiers. Some studies suggest that cryptocurrencies may offer hedging benefits against stock market downturns and could reduce portfolio risk during financial market turmoil (Gil-Alana et al. 2020; Pal and Mitra 2019; Al-Yahyaee et al. 2019; Jiang et al. 2021; Yousfi et al. 2023; Ustaoglu 2022; Xu and Kinkyo 2023). However, the high volatility of Bitcoin presents challenges for risk management, raising questions about its suitability as a safe-haven asset (Bouri et al. 2017; Shahzad et al. 2019; Chemkha et al. 2021). Overall, existing studies provide inconclusive evidence regarding the role of cryptocurrencies, whether they can truly serve as safe havens or merely as diversifiers. For example, Antonakakis et al. (2020) examine transmission mechanisms in the cryptocurrency market via the TVP-FAVAR connectedness approach, with a focus on nine high-market-capitalization cryptocurrencies. They reveal significant dynamic variability in total connectedness across cryptocurrencies, ranging between 25 and 75%, with high market uncertainty associated with strong connectedness and low uncertainty associated with weak connectedness. Kurka (2019) investigates spillovers among the cryptocurrency, commodity, foreign exchange, and stock markets, finding limited connectedness between cryptocurrencies and other markets, except for gold. Jiang et al. (2021) examine the hedging role of major cryptocurrencies against the MSCI and developed stock markets via quantile and frequency connectedness and quantile coherency methods, concluding that Ether has the highest hedging effectiveness against stock market shocks. Charfeddine et al. (2020) focus on the dynamic relationships among cryptocurrencies, major financial securities, and commodities, highlighting weak cross-correlations with traditional assets and suggesting cryptocurrencies as effective tools for financial diversification. Mezghani et al. (2021) explored the dynamic connectedness between stock markets and commodity futures during the 2014 oil price drop and the COVID-19 pandemic via the BEKK-GARCH model and Diebold and Yilmaz's methodology, underscoring the vulnerability of the oil market and identifying gold as an effective hedging tool during periods of turmoil. Ghorbel et al. (2022) explore return connectedness among cryptocurrencies, G7 stock indices, and gold during the COVID-19 pandemic, revealing a greater degree of independence among stock markets during the crisis, with cryptocurrencies generally acting as diversifiers for stock index portfolios, thereby reducing volatility. Additionally, Bitcoin has consistently been the largest net transmitter of volatility connectedness during the crisis, whereas gold has maintained its status as a net receiver, reinforcing its position as a safe-haven asset. Nguyen (2022) investigated the volatility connectedness between cryptocurrencies and the S&P 500 via VAR (1)-GARCH (1,1) and quantile regression, demonstrating volatility spillover from the S&P 500 to Bitcoin during financial turmoil. Zeng and Ahmed (2022) examine the effect of the COVID-19 outbreak on the volatility connectedness of China-ASEAN stock markets, cryptocurrencies, and crude oil markets via the GARCH-EVT-Vine-Copula and DCC-GARCH approaches, revealing stronger linkages between these markets during the pandemic than at prepandemic levels, with a notable shift in the Chinese equity and crude oil markets from being net transmitters to being net receivers of shocks. Balcilar et al. (2022) assess volatility connectedness between emerging equity markets and highly capitalized cryptocurrencies before and after the COVID-19 pandemic via time–frequency connectedness methodologies rooted in network analysis based on the quantile VAR model. They find that cryptocurrencies do not act as diversifiers for emerging stock markets in either the short or long term. Ha (2023) investigated the interconnections between cryptocurrency and stock markets during the COVID-19 pandemic, employing the TVP-VAR combined with an extended joint connectedness approach, revealing that pandemic shocks significantly influenced system-wide dynamic connectedness, peaking during the COVID-19 crisis. The findings indicate a heterogeneous role for each cryptocurrency and stock, with Bitcoin and Binance Coin emerging as net receivers of shocks, whereas Ethereum transitions from a receiver to a transmitter. Zhao and Zhang (2023) explore the dynamic return connectedness between cryptocurrency and global stock market uncertainty indices via TVP-VAR with stochastic volatility and conclude that cryptocurrency does not exhibit safe-haven properties against stock market shocks.

Several studies have examined the spillover effects of oil prices on the cryptocurrency market. Bouri et al. (2017) reported that prior to the 2013 Bitcoin crash, Bitcoin served as a high hedge and safe haven against energy commodities. Symitsi and Chalvatzis (2018) conclude that Bitcoin is less correlated with technology and energy indices, offering portfolio diversification opportunities. Additionally, Okorie and Lin (2020) show that Ethereum provides a hedge against crude oil in the short term, but Solve, Elastos, and Bit Capital Vendors do so in the long term. Using the Diebold and Yilmaz connectedness approach, Mandaci et al. (2020) studied the volatility spillover effect among global commodity futures and suggested that stock market volatility can be hedged by taking short positions in energy futures. Mezghani and Boujelbène-Abbes (2023) investigate the impact of financial stress on the correlation between oil and stock–bond markets in the GCC using the wavelet coherence model and time–frequency connectedness approach and find that the correlation remains stable during nonshock periods but fluctuates during oil and financial shocks at lower frequencies. They also suggest that investors adjust their portfolio compositions on the basis of the financial stress index to minimize risk during financial turmoil.

Some studies analyze how technological, economic, and geopolitical uncertainties affect the interconnectedness of cryptocurrency markets, shedding light on the factors shaping market dynamics. Research reveals the impact of cryptocurrency markets on the global technology sector, the dynamic interplay between Bitcoin and other cryptocurrencies, and the influence of cryptocurrency uncertainty indices on other financial asset classes during the pandemic. These findings provide insights into market dynamics and potential threats to financial stability. For example, Wajdi et al. (2020) study the contagion effect and dynamic interplay between Bitcoin and a broad spectrum of other cryptocurrencies, suggesting a dynamic conditional correlation and spillover between Bitcoin and its counterparts and emphasizing the importance of financial investors frequently adjusting their portfolios in response to specific cryptocurrency types and prevailing market conditions. Umar et al. (2021a) investigate the relationship between the technology sector and the cryptocurrency market via network connectedness measures developed by Diebold and Yilmaz, revealing that the impact of the cryptocurrency market on the global technology sector is relatively minor, indicating lower exposure to systemic risk. Umar et al. (2021b) explore the effects of COVID-19-related media coverage on the return and volatility connectedness of major cryptocurrencies (Bitcoin, Ethereum, Ripple) and fiat currencies (euro, GBP, and Chinese yuan) and find that cryptocurrencies and media coverage (before the first wave) act as net transmitters of shocks, whereas fiat currencies are generally net receivers. An exception is the euro, which transitioned from a net receiver to a net transmitter during the first wave of the pandemic, reflecting the impact of the virus on the European continent. Asiri et al. (2023) examine the dynamic relationship between cryptocurrency uncertainty indices and movements in returns and volatility across various financial assets, including cryptocurrencies, precious metals, green bonds, and soft commodities. Their findings suggest that cryptocurrency uncertainty indices significantly influence other financial asset classes during the pandemic, posing a potential threat to financial stability.

Numerous studies compare the characteristics of Bitcoin and gold. Some researchers find that Bitcoin possesses hedging capabilities and serves as an advantageous exchange medium, whereas others conclude that it does not qualify as a safe-haven asset. Gold, on the other hand, is often seen as offering superior diversification benefits and more effective hedging than Bitcoin in certain scenarios. For instance, Dyhrberg (2016) noted similarities between Bitcoin, gold, and the US dollar in terms of financial attributes, suggesting that Bitcoin represents hedging properties and offers opportunities as an exchange medium. Conversely, Klein et al. (2018) demonstrate that Bitcoin behaves differently from gold, particularly during market distress, indicating that it is not a safe-haven asset in developed markets. Selmi et al. (2018) compare the hedging and safe haven characteristics of Bitcoin and gold and find that both assets have hedging properties against oil price shocks. Smales (2019) shows that Bitcoin does not act as a safe haven for other assets. Shahzad et al. (2020) indicate that gold serves as a safe haven and hedge for G7 stock indices, whereas Bitcoin exhibits these characteristics only for Canada. Compared with Bitcoin, gold offers superior diversification advantages and more effective out-of-sample hedging. Elsayed et al. (2022) study the volatility connectedness of cryptocurrency, gold, and uncertainty indices and conclude that cryptocurrency policy uncertainty primarily drives return spillovers, with gold failing to provide a hedge against cryptocurrency uncertainty. Nakagawa and Sakemoto (2022) explore the relationship between connectedness from Bitcoin to gold and find a positive association between the expected returns of Bitcoin and gold. Mo et al. (2022) investigated the time and frequency connectedness between cryptocurrencies and commodity sectors and suggested that cryptocurrencies contribute to the system as the primary transmitter of risk spillover in both the short and long term. The analysis of the portfolio indicates that the hedging effect of cryptocurrencies differed among commodity sectors before and after COVID-19. Post-COVID-19, cryptocurrencies appear to be better hedging tools within the system.

The limited studies available indicate that research on the connections between cryptocurrencies, equities, and commodities presents a variety of perspectives and findings in portfolio management. Zeng et al. (2020) explore the dynamic connectedness of returns among Bitcoin, stocks, oil, and gold, revealing a weak linkage between Bitcoin and conventional assets, with an asymmetrical pattern evident in spillover effects, particularly when differentiating between positive and negative returns in the Bitcoin market. A rolling window analysis indicates a moderate increase in Bitcoin's connection to other financial assets. Bouri et al. (2021b) investigated the impact of the COVID-19 outbreak on return connectedness across various asset classes, including gold, crude oil, world equities, currencies, and bonds; employed the TVP-VAR connectedness approach; and revealed a significant shift in the structure and time-varying patterns of return connectedness around the onset of the pandemic. Ha and Nham (2022) analyze spillovers among cryptocurrency, commodity, and stock markets via the TVP-VAR with an extended joint connectedness approach and find that return connectedness from Bitcoin and Ethereum to the stock market increased during the COVID-19 outbreak. Ha and Nham (2022) further analyze the interconnections among commodity, stock, and cryptocurrency markets utilizing the TVP-VAR combined with an extended joint connectedness approach and find that the cryptocurrency market consistently acts as a time-varying net receiver and net transmitter, playing a relatively minor role. They also state that crude oil and stocks generally receive spillover effects from other markets, whereas the role of gold as a net transmitter or net receiver depends on the specific market under consideration. Xu and Kinkyo (2023) show that Bitcoin plays a dominant role over gold in terms of hedging effectiveness for the G7 stock markets, particularly during the Russia–Ukraine war period. Hanif et al. (2023) analyze high-order moments' connectedness among the cryptocurrency, commodity, and U.S., U.K., Eurozone, and Japanese stock markets via time and frequency connectedness on the basis of Diebold and Yilmaz (2012) and Barunik and Křehlík (2018). They find that cryptocurrency, stock, and commodity markets are closely connected in terms of volatility and volatility jumps. While skewness and kurtosis have fewer connections, volatility and jumps are more persistent. The level of connectedness varies over time, increasing during periods of uncertainty. Cagli and Mandaci (2023) examine how uncertainty in cryptocurrency, stock, currency, and commodity markets is connected via the time and frequency measures of connectedness developed by Diebold and Yilmaz (2012) and Barunik and Křehlík (2018). They demonstrate a low degree of uncertainty connectedness between cryptocurrency and other markets, suggesting long-term diversification opportunities and highlighting the unique dynamics of cryptocurrency markets. Echaust et al. (2024) examine whether hedging strategies using cryptocurrencies, gold, and oil can outperform index futures in mitigating stock market risk through multivariate GARCH models and conclude that from a risk-minimization perspective, there is no compelling reason for investors to incorporate cryptocurrencies, gold, or oil into hedging strategies against stock market risk.

The literature on the interconnectedness between cryptocurrencies and various financial assets, particularly in the context of the COVID-19 pandemic, reveals significant shifts in the patterns of return and volatility connectedness across different asset classes. These studies suggest that traditional diversification benefits may be challenged during periods of heightened uncertainty and market turbulence, emphasizing the evolving role of cryptocurrencies as both transmitters and receivers of risk. The increasing linkages within the cryptocurrency market and between digital and traditional assets highlight the need for continuous monitoring of these dynamic relationships, particularly in times of global crisis. This growing body of research offers valuable insights that can inform the strategies of investors and policymakers, stressing the importance of a more nuanced approach to understanding the complexities of financial markets amid ongoing global challenges.

Data

We examine the time-varying connectedness among the cryptocurrency, commodity, bond, and stock markets in Germany, India, and the United States, using daily return data from April 19, 2017, to December 29, 2023. The analysis begins in April 2017 to minimize experimental error, as the cryptocurrency market was in a crash phase before this period, leading to a substantial decline in market capitalization for most cryptocurrencies (Kumar et al. 2022; Huang 2024). Focusing on the most recent 6 years of data allows us to capture a period marked by rapid growth and heightened volatility in the cryptocurrency market, as well as significant economic and geopolitical events such as the U.S.–China trade war, the COVID-19 pandemic, the Russia–Ukraine conflict, and the Silicon Valley Bank collapse. These developments have profoundly influenced global financial markets, underscoring the necessity of investigating time-varying and frequency connectedness between cryptocurrencies and other markets during these uncertainties (Naeem et al. 2021; Xiao et al. 2021; Hanif et al. 2023). This approach ensures that the analysis reflects key market developments, providing valuable insights into market behavior during this dynamic period.

We further explore the inclusion of cryptocurrencies in portfolios, particularly their roles as safe-haven and hedging assets, as discussed in various studies (Guesmi 2011; Naeem et al. 2020; Zeng et al. 2020; Majdoub et al. 2021; Umar et al. 2021a,b; Ha and Nham 2022). Research has increasingly highlighted that investors often prefer cryptocurrencies as safe-haven assets, sometimes even over gold (Bouri et al. 2019; Jareño et al. 2020; Yousaf and Yarovaya 2022), and that they play significant roles in hedging and diversification during uncertain times (Yousaf and Ali 2020; Bakry et al. 2021; Joshi et al. 2022). The cryptoecosystem has experienced considerable growth in recent years, reshaping the global financial landscape. Bitcoin, known as “liquid gold” because of its independence from sovereign influences, remains the first cryptocurrency (Katsiampa et al. 2022). Its rise has led to the emergence of many other cryptocurrencies, which are vital for investment portfolios (Mo et al. 2022). Ethereum, the second-largest digital currency by market capitalization, has gained popularity for its unique features, including compatibility with nonfungible tokens (NFTs) that signify ownership of unique items secured on its blockchain (BenSaïda 2023). Additionally, stablecoins have become key tools for facilitating cryptocurrency, with Tether (USDT) being prominent owing to its high market capitalization and adoption rates (MacDonald and Zhao 2022). Tether's decentralized mechanism leverages market makers and arbitrageurs to exploit price differences, impacting liquidity and investor sentiment in the broader cryptocurrency market (Lyons and Viswanath-Natraj 2023). Bitcoin, Ethereum, and Tether (a stablecoin) have substantial market capitalization and trading volume (Echaust et al. 2024; Hsu et al. 2024). Therefore, we select Bitcoin, Ethereum, and Tether (a stablecoin) for analysis. Additionally, the inclusion of the U.S. Treasury 5-year bond in the portfolio is supported by the assumption that cryptocurrencies act as diversifiers owing to their weak correlation with bonds. Gold and crude oil are also commonly regarded as safe-haven assets by investors (Baur et al. 2015; Brière et al. 2015; Huang 2024). The DAX30, S&P 500, and SENSEX indices are considered because of their high trading volumes in both developed and emerging stock markets (Ghorbel et al. 2022; Sharma 2024). India is increasingly seen as a promising destination for foreign investors looking to diversify their portfolios (Aggarwal and Saradhi 2024). Additionally, the United States and Germany, as the largest industrialized economies, attract significant business investments. These markets experience intense cryptocurrency trading and mining activities (Küfeoğlu and Özkuran 2019; Cagli and Mandaci 2023). Analyzing country-specific stock indices allows for a comprehensive examination of how cryptocurrency covolatility dynamics interact globally. The distinct economic conditions, regulatory landscapes, and market behaviors in each country influence the relationship between cryptocurrencies and traditional assets. By including these indices, the study reveals how regional factors impact correlation and comovement trends, highlighting the importance of these countries in the global cryptocurrency landscape (Velappan 2024).

To address the asynchronous nature of stock market transactions and reduce the effects of nonsynchronous trading, we employed methods from Cai et al. (2009) and Zhou et al. (2012). Given that Indian markets open 8–12 h earlier than those in the U.S. and Germany, we lagged the Indian market's conditional return, volatilities, skewness, and kurtosis by one period in our VAR estimation (He and Hamori 2021).

Our analysis covers higher moment portfolios to illustrate the connectedness between cryptocurrencies and other markets. To measure higher moments, we utilize the GARCHSK model, appending the terms "price", "h", "sk", and "ku" to the variables to denote return, conditional volatility, conditional skewness, and conditional kurtosis, respectively. Table 1 provides a detailed description of the data used in our analysis.

Figures 1 and 2 present the historical price indices and returns of cryptocurrencies, including those of developed and developing stock markets, gold, crude oil, and 5-year U.S. Treasury bonds, covering the period from April 19, 2017, to December 29, 2023. Notably, the prices and returns of Bitcoin and Ethereum experienced a sharp decline in March 2020, triggered by the uncertainty surrounding the pandemic. However, by June 2022, both Bitcoin and Ethereum had reached their highest prices since December 2020. The subsequent decline in the cryptocurrency market is largely attributed to inflation data released by the Federal Reserve Bank (FED) in May 2022. Throughout 2023, growing interest from institutional investors further strengthened

Table 1 Definition of variables in our analysis

Abbreviation	Data definition
DAX	The closing price of the DAX index in U.S. dollars, a stock market index representing 40 major German companies trading on the Frankfurt Stock Exchange
SENSEX	This refers to the closing price of the SENSEX index in U.S. dollars, which is a stock market index representing 30 major companies listed on the Bombay Stock Exchange in India
S&P500	The closing price of the S&P 500 index in U.S. dollars, a stock market index representing 500 large companies listed on stock exchanges in the United States
ETH	The closing price of Ethereum (ETH), a cryptocurrency and a decentralized computing platform
BTC	The closing price of Bitcoin (BTC), the first and most well-known cryptocurrency
USDT	The closing price of Tether (USDT), a stablecoin cryptocurrency pegged to the value of the US dollar
BRENT	Crude oil (WTI) price in U.S. dollars
GOLD	Gold price by the ounce in U.S. dollars
US5BOND	The 5-year U.S. treasury bonds

Traditional financial data are imported from Yahoo Finance, whereas cryptocurrency market data are collected from coinmarketcap.com, Yahoo Finance, and investing.com

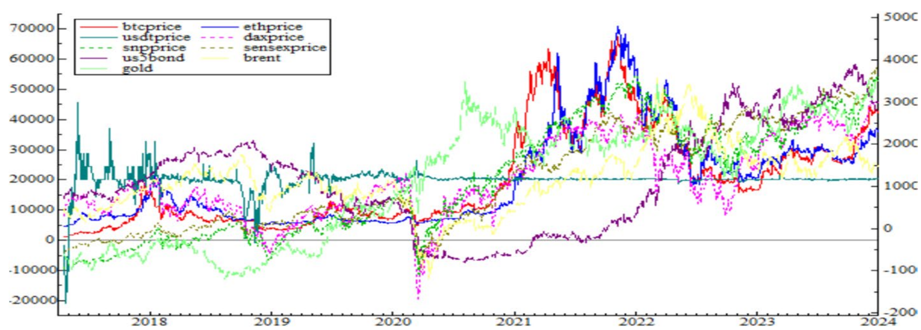


Fig. 1 Closing plots of the cryptocurrency, commodity, bond, and stock markets. *Note*: This plot illustrates the historical price indices of cryptocurrencies, developed and developing stock markets, gold, crude oil, and 5-year Treasury bonds from April 19, 2017, to December 29, 2023

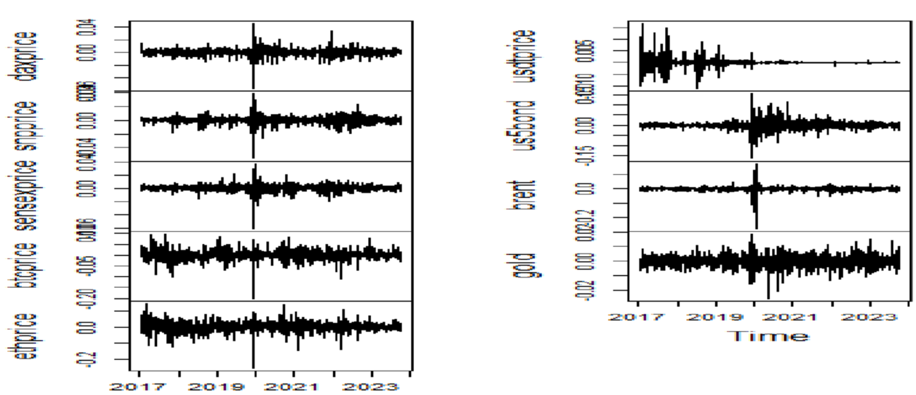


Fig. 2 Return plots of the cryptocurrency, commodity, bond, and stock markets. *Note*: This plot depicts the returns of cryptocurrencies, developed and developing stock markets, gold, crude oil, and 5-year Treasury bonds from April 19, 2017, to December 29, 2023. Returns are calculated via a log-difference transformation

Table 2 Descriptive statistics regarding returns of cryptocurrency and traditional financial markets

	Mean	Median	Maximum	Minimum	Std. Dev.	Skewness	Kurtosis	Jarque–Bera
Daxprice	8.50E−05	0.000281	0.045229	−0.0567	0.005321	−0.68009	17.19216	14,431.99
Snpprice	0.000181	0.000259	0.038949	−0.05544	0.005338	−0.87615	18.79718	17,936.12
Sensexprice	0.00023	0.000204	0.037326	−0.06124	0.004827	−1.5449	27.43792	43,079.86
Btcprice	0.000904	0.000706	0.097768	−0.20183	0.019972	−0.66222	12.50535	6539.518
Usdtprice	1.46E−05	0	0.016706	−0.01111	0.001717	1.309986	25.01122	34,886.43
Ethprice	0.000964	0.000363	0.142914	−0.25608	0.026706	−0.57219	11.52685	5255.189
Brent	9.94E−05	0.000686	0.178939	−0.27956	0.01497	−3.11265	95.88493	615,311.9
Gold	0.00012	0.000216	0.018661	−0.02561	0.003741	−0.29286	6.55952	923.9402
Us5bond	0.000207	0	0.155711	−0.16355	0.018353	−0.02667	19.0754	18,347.91

Table 3 Nonlinearity tests

	Terasvirta	White	Keenan	Tsay	LR
Panel A: Returns					
Daxprice	0.7701 (0.6804)	0.2676 (0.8747)	2.4868 (0.1149)	5.4538 (0.0000)	30.1756 (0.0000)
Snpprice	84.467 (0.0000)	89.071 (0.0000)	12.3249 (0.0000)	8.4612 (0.0000)	100.4407 (0.0000)
Sensexprice	5.9819 (0.0502)	10.469 (0.0053)	2.6175 (0.1058)	4.0614 (0.0000)	75.5089 (0.0000)
Btcprice	12.11 (0.0000)	15.21 (0.0004)	0.1405 (0.7078)	10.6587 (0.0011)	156.1483 (0.0000)
Usdtprice	25.647 (0.0000)	22.328 (0.0000)	0.1411 (0.7071)	5.1704 (0.0000)	35.9867 (0.0000)
Ethprice	26.041 (0.0000)	20.296 (0.0000)	1.8557 (0.1732)	30.2019 (0.0000)	200.6212 (0.0000)
Brent	159.67 (0.0000)	1.8507 (0.3964)	1.3515 (0.2451)	14.7921 (0.0000)	419.0115 (0.0000)
Gold	0.6661 (0.7167)	6.2848 (0.0431)	0.8964 (0.3438)	0.2785 (0.5976)	150.361 (0.0000)
Us5bond	33.1068 (0.0000)	2.0851 (0.3525)	9.1423 (0.0025)	8.4367 (0.0000)	64.6627 (0.0000)

the cryptocurrency market's ties with the traditional financial system. The small-scale banking crisis in the U.S. in March also brought renewed attention to the cryptocurrency market, where it emerged as a potential safe-haven asset amid the turmoil that led several medium-sized U.S. banks to bankruptcy. During this period, a significant increase of more than 20% was observed in March. Furthermore, the strong positive correlation of Bitcoin and Ethereum with the DAX Index, S&P 500 Index, and SENSEX Index, as illustrated in the figures, underscores their role as leading indicators for these cryptocurrencies.

Table 2 provides descriptive statistics for return changes across all markets, revealing key insights. Ethereum and Bitcoin exhibit the highest returns, but they also demonstrate greater volatility than the DAX, SENSEX, and S&P 500 indices do, suggesting that these cryptocurrencies are associated with greater risk. Moreover, skewness statistics indicate that all assets, with the exception of USDT, are left skewed. The kurtosis values for all assets exceed 3, suggesting that the return series exhibit leptokurtic distributions, which are characteristic of heavier tails and more extreme outliers than a normal distribution.

To investigate the potential nonlinear structure in return and volume changes, we apply several nonlinearity tests, including the Terasvirta, White, Keenan, Tsay, and likelihood ratio (LR) tests. These tests compare the null hypothesis of linearity against the alternative hypothesis of nonlinearity. As shown in Table 3, the null hypothesis is rejected for all return changes, indicating that the data exhibit nonlinear behavior.

Table 4 presents the results of the KSS unit root test based on the exponential smooth transition autoregressive (ESTAR) process. This test assesses the null hypothesis of nonstationarity against the alternative hypothesis of global stationarity. As indicated in Table 4, the rejection of the null hypothesis for the variables suggests that they follow a globally stationary ESTAR process.

Methodology

Higher moments measure

The examination of asset returns reveals a reality where they often exhibit nonnormal, asymmetric, and fat-tailed distributions. This highlights the necessity of considering higher-order moments of these distributions, extending beyond the traditional focus on return and volatility. By modeling the connectedness between cryptocurrencies and traditional financial markets through these higher-order moments, investors and portfolio managers can gain invaluable insights that aid in the construction of optimal portfolios. Higher-order moment risk connectedness can enhance both the effectiveness of hedging strategies and the overall cumulative returns within cryptocurrency-traditional financial asset portfolios. Consequently, investors may need to adjust their portfolio compositions frequently in response to fluctuations in these higher-order moments, employing various hedging and risk-averse strategies to effectively navigate changing market conditions. Additionally, the time-varying nature of these higher-order moments can act as warning signals, illuminating evolving risk patterns and providing real-time insights into the impacts of economic and geopolitical factors on the market. Policymakers could also adapt their strategies in real time on the basis of fluctuations in higher-order moments, potentially mitigating instability across various markets, including cryptocurrencies, stocks, bonds, crude oil, and gold (Bouri et al. 2021a, b, c, d).

The motivation for this analysis lies in the understanding that asset return distributions often deviate from normality and are characterized by an excess of extreme outcomes (Fama 1965). As noted by Rubinstein (1973), nonnormal distributions in asset returns, coupled with nonquadratic investor utility functions, compel investors to consider not only the mean and variance but also higher-order moments such as skewness

Table 4 KSS unit root test results for all markets

	Test statistics	Critical value
Daxprice	−3.7219 (6)	−2.93
Snpprice	−9.9188 (6)	−2.93
Sensexprice	−6.3210 (7)	−2.93
Btcprice	−5.8824 (7)	−2.93
Usdtprice	−11.1423 (7)	−2.93
Ethprice	−6.8454 (7)	−2.93
Brent	−15.7964 (6)	−2.93
Gold	−8.7028 (2)	−2.93
Us5bond	−7.2056 (5)	−2.93

The values in the parentheses are optimal lag lengths based on Akaike information criteria. −2.93 indicates the critical value at a 5% significance level and kurtosis. These higher-order moments provide crucial insights into investor preferences concerning risk and return. Specifically, skewness (the third moment) indicates the likelihood of market crashes, whereas kurtosis (the fourth moment) signals the probability of extreme events characterized by fat tails or leptokurtosis (Silva da Gama et al. 2019; Cui and Maghyereh 2023; León et al. 2005). To analyze these dynamics, the study utilizes the GARCHSK model following Gao et al. (2024) and Cui et al. (2024) to derive sequences of conditional volatility, skewness, and kurtosis for the return series of the cryptocurrency, stock, bond, crude oil, and gold markets, constructing a comprehensive higher-order moment fluctuation model.

$$
\begin{aligned}
r_t &= \mu_t + e_t = \mu_t + h_t^{1/2} Z_t \\
h_t &= \beta_0 + \sum_{i=1}^{q_t} \beta_{1,i} \varepsilon_{t-i}^2 + \sum_{j=1}^{p_t} \beta_{2,t} h_{t-j} \\
s_t &= \gamma_0 + \sum_{i=1}^{q_t} \gamma_{1,i} z_{t-i}^{p_z} + \sum_{j=1}^{p_t} \gamma_{2,j} s_{t-j} \\
k_t &= \delta_0 + \sum_{i=1}^{q_t} \delta_{1,i} z_{t-i}^4 + \sum_{j=1}^{p_t} \delta_{2,j} k_{t-j}
\end{aligned}
$$

$\varepsilon_t \big| I_{t-1} \sim D(0, h_t, s_t, k_t), I_{t-1}$ refers to the information set at time t − 1. $D(0, h_t, s_t, k_t)$ indicates the Gram–Charlier distribution, which involves the conditional mean, variance, skewness, and kurtosis. h_t and s_t, k_t represent the conditional variance, conditional skewness, and conditional kurtosis, respectively. r_t indicates the return vector for the cryptocurrency, stock, bond, crude oil, and gold markets.

TVP-VAR model with stochastic volatility

We utilize the time-varying parameter vector autoregression (TVP-VAR) connectedness method, as introduced by Antonakakis et al. (2020), to explore the interconnectedness and spillover effects between cryptocurrencies and traditional financial markets while also considering the impact of uncertainty in so-called markets. We prefer the TVP-VAR framework over models such as the BEKK-GARCH, DCC-GARCH, and VAR-GARCH because of its significant advantages. These alternative models have notable drawbacks, including (a) their inability to estimate time-varying spillovers and (b) challenges in estimation that can arise from convergence issues (Yousaf and Ali 2020; Yousaf et al. 2025). The TVP-VAR framework effectively addresses these limitations and allows variances to change over time via Kalman filter estimation with decay factors. This method reduces the issues associated with selecting rolling window sizes, which can lead to erratic parameters and the loss of valuable data (Antonakakis and Gabauer 2017; Antonakakis et al. 2018; Korobilis and Yilmaz 2018). Additionally, TVP-VAR avoids the loss of important observations found in the methods proposed by Diebold and Yilmaz (2012, 2014) owing to rolling windows (Korobilis and Yilmaz 2018; Yousaf and Yarovaya 2022; Yousaf et al. 2025).

This approach offers a comprehensive tool for capturing dynamic relationships over time. This study favors the TVP-VAR model over alternative models for several key reasons. First, the TVP-VAR model features a time-varying covariance structure that allows us to investigate dynamic interactions between variables. This means that we can assess how relationships may change over time, making it particularly well suited for analyzing connectedness in financial markets. Second, the TVP-VAR framework eliminates the need to arbitrarily define a rolling window size or address the loss of observations, as no rolling window analysis is needed. Additionally, it employs a multivariate Kalman filter, which is less sensitive to outliers, enhancing the robustness of our findings. Third, the TVP-VAR model provides considerable flexibility, allowing us to incorporate time-varying structures into both the autoregressive parameters and the variance–covariance matrix of the errors. This flexibility is vital because financial data often exhibit changes in structure, regime shifts, and periods of increased volatility. Most importantly, following the work of Broadstock et al. (2022), we use the estimated time-varying variance–covariance matrix from the TVP-VAR model to construct various portfolio strategies. In a standard VAR model, the coefficients and error covariance matrix remain fixed over time, an assumption that can become unrealistic during periods of economic turbulence or regime shifts, such as financial crises or pandemics. The TVP-VAR model with stochastic volatility allows both the parameters and the covariance matrix to evolve, effectively capturing changing relationships and modeling volatility clustering. The presence of significant autocorrelation in the data indicates that both the mean and variance may vary over time, which makes the TVP-VAR model with a time-varying variance–covariance structure particularly well suited for these conditions. This flexibility significantly enhances forecasting accuracy, especially in volatile market environments characterized by structural breaks (Primiceri 2005). Moreover, permitting the variance–covariance matrix to change over time improves estimation performance through Kalman filter estimation, addressing issues related to the arbitrary selection of rolling window sizes (Korobilis and Yilmaz 2018). Time-varying covariances also lead to more accurate prediction intervals, which are essential for effective risk management and policy design. In volatile environments, point forecasts alone are not sufficient; understanding the full distribution of potential outcomes is crucial. Accounting for changing volatilities results in more informative posterior distributions, significantly improving the quality of probabilistic forecasts (Cogley and Sargent 2005). The unique advantages of the TVP-VAR connectedness approach include its ability to deliver results unaffected by the size of a rolling window and the elimination of observation loss. Furthermore, the TVP-VAR approach is capable of analyzing connectedness in low-frequency datasets, further enhancing its value as a powerful analytical tool (Xu et al. 2024).

The TVP-VAR model with stochastic volatility is particularly effective in estimating the evolving relationship between financial and macroeconomic indicators, providing greater flexibility and robustness than traditional VAR models (Primiceri 2005; Nakajima 2011). A key feature of this model is the assumption of a lower-triangular matrix for A_t, which serves as a form of recursive identification for the VAR system. For simplicity, the estimation algorithm presented here focuses on recursive identification. Unlike standard VAR models, the parameters in the TVP-VAR model are not assumed to follow a stationary process, such as AR(1), but instead follow a random walk process. Given the extensive number of parameters in the TVP-VAR model, reducing the parameter count by assuming a random walk process for parameter innovation is advisable. However, extending the estimation algorithm to accommodate a stationary process is feasible (Nakajima 2011).

$$y_t = B_{1t}y_{t-1} + \cdots + B_{kt}y_{t-k} + u_t, u_t \sim N\left(0, A_t^{-1}\sum_t \varepsilon_t\right), t = 1, \ldots, T \tag{1}$$

where y_t indicates the $nx1$ vector of variables and where $B_{it}, i = 1, \ldots, k$ indicates nxn time-varying coefficient matrices. u_t indicates the $nx1$ vector of structural shocks, which has a lower triangular matrix A_t and diagonal matrix $\sum_t. \varepsilon_t \sim N(0, 1)$.

$$A_t = \begin{bmatrix} 1 & 0 & \ldots & 0 \\ a_{21,t} & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ a_{n1,t} & \cdots & a_{n(n-1)t} & 1 \end{bmatrix} \sum_t = \begin{bmatrix} \sigma_{1t}^2 & 0 & \ldots & 0 \\ 0 & \sigma_{2t}^2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \sigma_{nt}^2 \end{bmatrix}$$

When $X_t = I_n \otimes \left(y_{t-1}, \ldots, y_{t-k}\right)$ is considered. $\otimes$ indicates the Kronecker product. Thus, the model can be reconstructed as follows:

$$y_t = X_t\beta_t + A_t^{-1}\sum_t \varepsilon_t, t = s + 1, \ldots, n \tag{2}$$

The coefficients of B_{it}, A_t, and $\sum_t$ change over time. $\beta_t = vec(B_t), a_t$ are the covariance vectors of the elements in A_t, where $a_t = \left(a_{21,t}, \ldots, a_{n(n-1)t}\right)$. σ_t^2 are the variance vectors of the diagonal elements in $\sum_t$, where $\sigma_t^2 = \left(\sigma_{1t}^2, \ldots, \sigma_{nt}^2\right)$. β_t, a_t, and σ_t^2 pursue a random walk process:

$$\beta_t = \beta_{t-1} + v_t, v_t \sim N\left(0, \sum_\beta\right) \tag{3}$$

$$a_t = a_{t-1} + \zeta_t, \zeta_t \sim N\left(0, \sum_a\right) \tag{4}$$

$$log(\sigma_t) = log(\sigma_{t-1}) + \eta_t, \eta_t \sim N\left(0, \sum_\sigma\right) \tag{5}$$

The parameters in the TVP-VAR model follow a joint normal distribution. The parameters are as follows:

$$\begin{bmatrix} \varepsilon_t \\ v_t \\ \zeta_t \\ \eta_t \end{bmatrix} = \begin{bmatrix} I_n & 0 & 0 & 0 \\ 0 & \sum_\beta & 0 & 0 \\ 0 & 0 & \sum_a & 0 \\ 0 & 0 & 0 & \sum_\sigma \end{bmatrix}$$

The TVP-VAR model evaluates high-dimensional time-varying parameters via the Bayesian approach, which is based on the Markov chain Monte Carlo (MCMC) algorithm, to estimate the parameters in the study (Zhao and Zhang 2023). In the TVP-VAR model with stochastic volatility, we use a Cholesky decomposition for the covariance matrix of $\sum_t$, as suggested by Lucey et al. (2022), to order the targeted series:

$$y_t = \left(daxprice, snpprice, sensexprice, btcprice, usdtprice, ethprice, brent, gold, us5bond\right)$$

Nakajima (2011) discussed the steps for drawing samples via MCMC procedures. On the basis of data y_t, we use the MCMC method to draw samples from the posterior distribution. The TVP-VAR model with stochastic volatility is estimated using 1000 presamples and 10,000 stimulated samples in the study. This approach helps in generating samples that can be used for further analysis and inference in the context of the TVP-VAR model with stochastic volatility. The sampler takes the form:

$$\begin{array}{l} \beta_t|a_t, \sigma_t, \sum_\beta \\ a_t|\beta_t, \sigma_t, \sum_a \\ \sigma_t|\beta_t, a_t, \sum_\sigma \\ \sum_\beta |\beta_t \\ \sum_a |a_t \\ \sum_\sigma |\sigma_t \end{array}$$

Dynamic connectedness ındex

The estimation of the generalized connectedness procedure by Diebold and Yilmaz (2012, 2014) relies on time-varying coefficients and variance–covariance matrices. This method is based on generalized impulse response functions (GIRFs) and generalized forecast error variance decompositions (GFEVDs) following the approaches of Koop et al. (1996) and Pesaran and Shin (1998). To compute the GIRF and GFEVD, we transform the TVP-VAR into its vector moving average (VMA) representation via the Wold representation theorem. The process of obtaining the VMA representation involves recursive substitution. The VMA representation is represented as $\sum_{j=0}^{\infty} A_{jt}\varepsilon_{t-j}$, where A_{jt} is the *nxn* matrix.

Supposing that $\left|\beta_{(z)}\right|$ are outside the unit circle, the VAR process is conventionally portrayed through the MA(∞) process as $y_t = \Psi(L)\varepsilon_t$, representing a fundamental aspect of time-varying relationship modeling in empirical macroeconomics. This representation holds significant implications for understanding and analyzing the dynamic connectedness index in the TVP-VAR model with stochastic volatility. $\Psi(L)$ indicates a matrix of infinite lag polynomials (Koop and Korobilis 2013).

The relationships between variables in such a system are generally demonstrated via forecast error variance decomposition, as shocks in one variable can affect others as well as itself. Since these shocks may not occur independently, standard Cholesky decomposition, which relies on ordering variables on the basis of the expected propagation of shocks, may not be suitable. To assess the pairwise directional connectedness from variable (j) to variable (i), we utilize the generalized forecast error variance decomposition (GFEVD) methodology proposed by Koop and Korobilis (2014). This method effectively illustrates the influence of variable (j) on variable (i) in terms of the forecast variance share. The GFEVD is represented as:

$$\Gamma_{ij,t}(K) = \frac{\sum_{t=1}^{K-1} \omega_{ij,t}^2}{\sum_{j=1}^{n} \sum_{t=1}^{K-1} \omega_{ij,t}^2} \tag{6}$$

$\sum_{j=1}^{n} \Gamma_{ij,t}(K) = 1$ and $\sum_{ij=1}^{n} \Gamma_{ij,t}(K) = n$. H indicates the number of forecasting horizons, and we select 20 days.

$GIRF\left(\Psi_{ij,t}(K)\right)$ is calculated as:

$$GIRF_t\left(H, p_{j,t}, \Omega_{t-1}\right) = E\left(y_t + H|e_j = p_{j,t}, \Omega_{t-1}\right) - E\left(y_{t+j}|\Omega_{t-1}\right) \tag{7}$$

$$\Psi_{ij}(K) = \frac{A_{H,t} \sum_t e_j}{\sqrt{\sum_{jj,t}}} \frac{p_{j,t}}{\sqrt{\sum_{jj,t}}} p_{j,t} = \sqrt{\sum_{jj,t}} \tag{8}$$

$$\Psi_{ij}(K) = \sum_{jj,t}^{-1/2} A_{H,t} \sum_{t} e_j \tag{9}$$

e_j indicates a nx1 vector whose jth element equals 1.

We aim to investigate whether having knowledge of cryptocurrency trends provides an economic advantage in the study. For this purpose, we research the dynamic return and volatility connectedness between cryptocurrency and other financial markets in terms of return and volatility, providing valuable insights into investment strategies and portfolio implications.

Analyzing networks is crucial in regard to understanding financial markets. The total connectedness index (TCI) is a measure of the spillover effect of a shock in one variable to others. On the basis of Monte Carlo simulations in Antonakakis et al. (2020), own variance shares are always larger than or equal to cross variance shares, which constrains the TCI to be within the range of $\left[0, \frac{k-1}{k}\right]$. However, since we want to know the average amount of network comovement as a percentage, which should fall between 0 and 1, a slight adjustment to the TCI is necessary. To determine the average level of comovement between networks as a percentage, the TCI needs to be slightly adjusted. The TCI is calculated as follows (Broadstock et al. 2022):

$$TCI_t^g(K) = \frac{\sum_{i,j=1, i \neq j}^{m} \vartheta_{ij,t}^q(K)}{k-1}, \qquad 0 \leq TCI_t^g(K) \leq 1 \tag{10}$$

To calculate the pairwise connectedness index (PCI) between assets i and j, the TCI index can be built as follows:

$$PCI_{ijt}(K) = 2\left(\frac{\vartheta_{ij,t}^q(K) + \vartheta_{ji,t}^q(K)}{\vartheta_{ii,t}^q(K) + \vartheta_{ij,t}^q(K) + \vartheta_{ji,t}^q(K) + \vartheta_{jj,t}^q(K)}\right), \qquad 0 \leq PCI_{ijt}^g(K) \leq 1 \tag{11}$$

The total directional connectedness from all others is calculated as follows:

$$\Gamma_{i \leftarrow j,t}(K) = \frac{\sum_{j=1, i \neq j}^{m} \vartheta_{ij,t}(K)}{\sum_{i=1}^{m} \vartheta_{ij,t}(K)} * 100 \tag{12}$$

Since the influence of variable i on itself has been excluded, the total directional connectedness from all the other variables on variable i must be strictly less than 100%.

Considering the GFEVD, the total directional connectedness to all others is calculated as follows:

$$\Gamma_{i\to j,t}(K) = \frac{\sum_{j=1,i\neq j}^{n} \vartheta_{ji,t}(K)}{\sum_{j=1}^{n} \vartheta_{ji,t}(K)} * 100 \tag{13}$$

This measure calculates the influence that variable i has on all the other variables j in the system by accumulating the effects (error variance) that variable i has on each of the other variables' forecast error variance. Moreover, the total directional connectedness measures can take values below, equal to, or above 100%.

The net total directional connectedness defines the comprehensive influence of a variable i on the entire network. This measure takes into account the cumulative impact of variable i's forecast error variance on each of the other variables in the system, providing valuable insights into the overall significance of the variable within the network. The net total directional connectedness is calculated as follows:

$$\Gamma_{i,t} = \Gamma_{i\to j,t}(K) - \Gamma_{i\leftarrow j,t}(K) \tag{14}$$

If $\Gamma_{i,t} > 0$, variable i has a significant influence on the network. Conversely, if $\Gamma_{i,t} < 0$, variable i is influenced by the network. The net pairwise directional connectedness is a valuable tool for analyzing the bidirectional relationships between variable i and variable j:

$$NPDC_{ij}(K) = \left(\vartheta_{ji,t}(K) - \vartheta_{ij,t}(K)\right) * 100 \tag{15}$$

The net pairwise directional connectedness (NPDC) measures the dominance between variables i and j. If NPDC > 0, i dominates j; if NPDC < 0, j dominates i. This quantifies the net effect of directional connectedness, capturing the difference between gross shocks transmitted from i to j and vice versa (Antonakakis et al. 2020).

In a secondary analysis, we explore various portfolio model specifications, which are the minimum variance portfolio (MVP), minimum correlation portfolio (MCP), and pairwise minimum connectedness portfolio (MCoP) approaches, which are based on investment objectives, risk tolerance, and market conditions. Our initial step involves applying the MVP approach (Durand 1960) for calculating portfolio weights:

$$w_t = \frac{H_t^{-1} I}{I H_t^{-1} I} \tag{16}$$

where w_t indicates the portfolio weight vector and where H_t defines the conditional variance–covariance matrix.

Our second portfolio is the MCP, proposed by Christoffersen et al. (2014). In this approach, portfolio weights are determined by the conditional correlation matrix rather than the conditional variance–covariance matrix. Before constructing this multivariate portfolio, we articulate the conditional correlations in the equation as follows:

$$R_t = diag(H_t)^{-0.5} H_t diag(H_t)^{-0.5}$$

R_t indicates a mxm-dimensional matrix. Based on R_t, the portfolio weights for each asset in the MCP are calculated as follows:

$$w_t = \frac{R_t^{-1} I}{I R_t^{-1} I} \tag{17}$$

The MCoP approach, which depends on the minimum pairwise interconnectedness among all assets and was proposed by Broadstock et al. (2022), creates a portfolio that proposes minimizing bilateral interdependence among variables. This portfolio strategically assigns higher weights to assets with low connectedness to others, effectively minimizing spillover effects. This approach not only reduces risk but also safeguards overall stability, making it a smart choice for resilient investing. The MCoP is calculated as follows:

$$w_t = \frac{PCI_t^{-1} I}{I PCI_t^{-1} I} \tag{18}$$

where PCI indicates the pairwise connectedness index matrix. I denotes the identity matrix.

We also aim to test the hedging performance of each asset in the portfolio. For this purpose, we employ the hedge effectiveness (HE) approach of Ederington (1979). The hedge effectiveness is calculated as follows:

$$HE = 1 - \frac{Var\left(y_p\right)}{Var\left(y_{unhedged}\right)} \tag{19}$$

$Var\left(y_p\right)$ indicates the variance of returns in the portfolio, but $Var\left(y_{unhedged}\right)$ is the variance of the unhedged asset.

Sensitivity analysis to the choice of prior

The TVP-VAR model, with its numerous state variables modeled as a nonstationary process, requires carefully selected priors to ensure accurate results. The model's flexibility allows it to capture both sudden and gradual changes in underlying economic conditions. As noted by Primiceri (2005), applying a tight prior to the covariance matrix of the disturbance in the random walk process can help prevent implausible behaviors in the time-varying parameters. However, caution is necessary when using tighter priors in empirical econometrics because of their inherent arbitrariness, although slightly tighter priors are often necessary for reasonable identification in the TVP-VAR model (Nakajima et al. 2011).

Although the VAR model can effectively obtain both sudden and gradual changes in the underlying economic conditions, introducing time variation in every parameter may lead to an overidentification problem. To address this issue, Primiceri (2005) suggested implementing a tight prior for the covariance matrix of the disturbance in the random walk process to prevent implausible behaviors of the time-varying parameters. Notably, the time-varying coefficient β_t necessitates a tighter prior than the simultaneous relations a_t and the volatility σ_t of the structural shock for the variance of the disturbance in their time-varying process. The fluctuations in the size of the structural shock affecting the economic system are expected to vary more widely over time than the potential changes in the autoregressive system of the economic variables specified by VAR coefficients. In most of the relevant literature, a tighter prior is established for $\sum\beta$, whereas a more diffuse prior is set for $\sum a$ and $\sum\sigma$. Conducting a prior sensitivity analysis is crucial to ensure the robustness of the empirical results with prior tightness (Nakajima 2011).

Certainly, it is crucial to establish the prior initial state of the time-varying parameters, particularly when dealing with a stationary time series model. Typically, we assume that the initial state follows a stationary distribution of the process. To establish a specific prior for β_{s+1}, a_{s+1}, and σ_{s+1}, we can adopt one of two approaches. The first approach, proposed by Primiceri (2005), entails setting a normal distribution prior, with the mean and variance determined on the basis of estimates from a constant parameter VAR model calculated using the presample period. This approach allows us to leverage the economic structure estimated from the presample period up to the initial period of the primary sample data. Alternatively, the second approach involves adopting a relatively flat prior for the initial state, considering our lack of prior information about it (Koop and Korobilis 2010; Korobilis and Yilmaz 2018).

We perform a sensitivity analysis to measure the impact of different prior assumptions on the results. In other words, we check whether the time series behavior of the connectedness index is due to the choice of the Bayesian TVP-VAR model we estimate, namely, the Minnesota prior. We analyze the time series behavior of the connectedness index on the basis of the Minnesota prior and an alternative using the Primiceri (2005) prior. We initially utilize the prior settings, known as Bayesian priors, used in Primiceri (2005) to estimate the TVP-VAR model. Bayesian methods help manage dense parameterization by incorporating prior information, addressing potential overparameterization issues from limited macroeconomic data (Koop and Korobilis 2010). Informative priors enhance the model structure and improve the forecasting performance (Koop 2013). This method uses a specific number of initial observations as the training sample. Bayesian inference often involves the use of MCMC methods such as the Gibbs sampler. In TVPVAR estimation, the Bayesian methodology offers advantages over frequentist methods by enabling the use of powerful computational algorithms suited for addressing time variation. Bayesian inference involves defining a prior distribution for the model's parameters, updating it with the data via a likelihood function, and obtaining the posterior distribution. In this context, estimators are statistics of this distribution, such as the mean or mode. In this study, we opted for a training sample of 200 daily observations regarding a trading year. As an alternative to the TVP-VAR model with Bayesian priors proposed by Primiceri (2005), we also estimated the TVP-VAR model using the Minnesota priors to exhibit the behavior of cryptocurrencies and other financial markets in the study. The Minnesota prior (Litterman 1980) assumes that variables follow random walk processes, typically yielding accurate forecasts of macroeconomic time series (Kilian and Lütkepohl 2017). This prior posits that all assets follow a univariate process, which can exhibit varying degrees of persistence, from highly persistent to a random walk. Refinements of the Minnesota prior are often implemented as additional priors to "reduce the importance of the deterministic component implied by VAR models estimated conditioning on the initial observations" (Giannone et al. 2015, p. 440). Given initial conditions and estimated coefficients, this component is defined as the expectation of future observations. Given that the parameters fall within the -0.99 to 0.99 range, no series can be nonstationary. The parameters are selected randomly, enabling us to explore different settings and observe the resulting variations in the analysis. Our findings are based on 500 randomly chosen Minnesota priors, following the approach of Antonakakis et al. (2020). Our findings remain robust across various estimations and prior choices. For much of the sample period, the TVP-VAR model based on the Primiceri prior closely mirrors the results from the Minnesota prior, confirming the robustness of our main findings.

Empirical findings

Static connectedness findings

To estimate the return and volatility connectedness among cryptocurrencies, gold, crude oil, the 5-year U.S. bond, and stock markets, we employ the TVP-VAR connectedness model introduced by Antonakakis et al. (2020), which is based on Diebold and Yilmaz's (2012, 2014) spillover index. This model uses a rolling window of 200 days, a lag length of order 1 (determined by the BIC), and a 20-step-ahead forecast. By doing so, we examine the roles of sender and receiver in each market, capturing the interconnections among all markets. The TVP-VAR model is particularly suited for this analysis, as it allows coefficients to quickly adapt to adverse events, such as economic and financial shocks, and to evolve over time (Pham and Do 2022). The concept of connectedness provides valuable insights into spillover transmission, with significant implications for both investors and policymakers.

Table 5 presents the static connectedness between cryptocurrencies and other financial markets for higher-order moments. Each value in Table 5 represents the percentage of forecast error variance in the row variable that is explained by the column variable. In this context, "TO" indicates the connectedness effects transmitted by each asset to the entire system, whereas "FROM" reflects the connectedness effects received by each asset from the system. The "NET" column measures the net connectedness index, where markets with a positive NET value act as shock transmitters and those with a negative NET value act as shock receivers. The total connectedness index (TCI) captures the overall degree of connectedness within the system.

Panel A shows that the mean return connectedness across markets is 40.08%, with variations observed over time. Bitcoin and Ethereum exhibit high levels of correlation, with cross-market connectedness indices of 30.83% and 30.76%, respectively. In contrast, the correlations among the three stock markets (DAX, S&P 500, and SENSEX) are relatively low. Similarly, the stock, commodity, and U.S. 5-year bond markets demonstrate low connectedness with cryptocurrency markets, with cross-market connectedness indices between cryptocurrencies and other markets ranging from 0.61% to 5.31%. The "FROM" and "TO" connectedness indices reveal that the stock markets (DAX and S&P 500) and cryptocurrency markets (Bitcoin and Ethereum) are the highest receivers and transmitters of shocks within the system. Finally, the NET index indicates that Bitcoin, Ethereum, DAX, and S&P 500 are net shock transmitters, whereas USDT, SENSEX, commodities, and the U.S. 5-year bond markets are net shock receivers.

Table 5 Total connectedness index between cryptocurrencies and other markets

	daxprice	snpprice	sensexprice	Btcprice	ethprice	usdtprice	Us5bond	brent	gold	FROM
Panel A. Return connectedness										
Daxprice	52.03	19.73	7.92	3.59	3.58	2.38	4.38	3.61	2.79	47.97
Snpprice	17.91	54.75	4.91	5.22	5.31	1.18	4.78	3.25	2.69	45.25
Sensexprice	12.02	12.22	62.86	2.43	3.15	1.51	1.84	2.45	1.52	37.14
Btcprice	3.23	4.94	1.2	54.1	30.83	1.86	0.86	1.05	1.92	45.9
Ethprice	3.22	5.09	1.87	30.76	54.04	1.54	0.74	1.16	1.58	45.96
Usdtprice	2.65	1.96	1.5	3.01	2.61	84.93	0.88	1.23	1.24	15.07
Us5bond	5.82	5.81	1.99	1	1.03	0.67	68.67	2.36	12.65	31.33
Brent	4.95	4.87	2.04	1.87	1.77	0.61	2.88	77.63	3.37	22.37
Gold	3.73	3.17	1.23	2.65	2.07	0.88	12.93	2.99	70.34	29.66
TO	53.52	57.8	22.67	50.52	50.36	10.63	29.29	18.1	27.77	320.66
NET	5.55	12.54	−14.47	4.63	4.4	−4.44	−2.05	−4.28	−1.89	**TCI**/40.08
	daxh	**snph**	**sensexh**	**btch**	**ethh**	**usdth**	**us5bondh**	**brenth**	**goldh**	**FROM**
Panel B. Volatility connectedness										
Daxh	50.01	14.71	6.59	4.16	4.32	1.16	8.05	6	4.99	49.99
Snph	8.76	61.74	3.61	5.42	4.94	2.44	5.9	3.58	3.62	38.26
Sensexh	7.88	10.79	57.9	2.55	3.13	2.92	6.24	4.32	4.27	42.1
Btch	2.49	5.99	1.87	52.63	26.68	3.18	2.4	2.27	2.49	47.37
Ethh	2.85	6.47	1.77	26.66	52.04	4.03	2.43	1.7	2.05	47.96
Usdth	5.19	4.13	4.7	6.59	8.28	52.8	6.07	7.03	5.21	47.2
Us5bondh	7.15	4.85	4.79	2.94	3.27	3.69	57.82	7.37	8.13	42.18
Brenth	7.49	2.78	3.74	1.41	1.74	1.39	8.88	66.38	6.18	33.62
Goldh	5.83	5.76	3.75	3	3.58	2.1	12.02	5.55	58.41	41.59
TO	47.64	55.49	30.82	52.73	55.94	20.9	51.99	37.83	36.93	390.26
NET	−2.35	17.23	−11.28	5.36	7.98	−26.3	9.81	4.21	−4.66	48.78

Table 5 (continued)

	daxsk	snpsk	sensexsk	btcsk	ethsk	usdtsk	us5bondsk	brentsk	Goldsk	FROM
Panel C. Skewness connectedness										
Daxsk	59.27	5.57	8.32	4.48	4.93	2.83	5.07	6.89	2.64	40.73
Snpsk	5.41	77.86	1.8	2.21	1.96	0.7	2.28	6.07	1.72	22.14
Sensexsk	10.22	2.1	63.36	4.29	4.3	2.08	2.96	8.85	1.84	36.64
Btcsk	3.9	2.56	3.23	51.52	25.44	5.22	2.25	3.75	2.12	48.48
Ethsk	4.41	1.87	3.76	26.65	51.16	6.81	1.73	2.05	1.55	48.84
Usdtsk	2.32	0.89	1.83	6.48	8.33	77.92	0.9	0.85	0.48	22.08
Us5bondsk	5.83	2.03	4.19	2.13	2.14	1.41	65.55	8.27	8.45	34.45
Brentsk	5.68	3.61	4.56	1.92	1.69	0.44	3.5	74.92	3.69	25.08
Goldsk	3.46	1.86	1.7	2.86	2.21	0.74	8.79	4.91	73.49	26.51
TO	41.22	20.49	29.4	51.01	51.01	20.23	27.46	41.64	22.49	304.95
NET	0.49	−1.66	−7.24	2.53	2.18	−1.85	−6.99	16.56	−4.02	38.12
	daxku	**snpku**	**sensexku**	**btcku**	**ethku**	**usdtku**	**us5bondu**	**brentku**	**Goldku**	**FROM**
Panel D. Kurtosis connectedness										
Daxku	32.24	7.99	9.06	4.35	5.4	8.74	12.52	15.21	4.49	67.76
Snpku	6.76	64.06	3.58	2.82	2.65	7.29	5.29	5.24	2.32	35.94
Sensexku	7.97	4.16	44.2	4.23	4.73	13.55	8.93	9.55	2.67	55.8
Btcku	5.18	4.72	4.68	38.85	17.01	13.69	5.87	6.91	3.08	61.15
Ethku	5.55	3.68	5.05	17.55	39.19	13.89	6.02	6.39	2.67	60.81
Usdtku	4.37	2.2	3.62	4	5.98	56.48	6.46	14.96	1.92	43.52
Us5bondku	4.77	3.69	5.59	3.16	3.33	13.8	44.31	15.83	5.53	55.69
Brentku	6.89	3.65	5.26	3.05	3.3	12.3	14.79	46.85	3.92	53.15
Goldku	3.36	2.77	2.83	3.44	3.04	6.63	10.64	7.41	59.88	40.12
TO	44.86	32.86	39.68	42.6	45.45	89.89	70.51	81.5	26.58	473.95
NET	−22.9	−3.08	−16.11	−18.55	−15.37	46.37	14.82	28.35	−13.54	59.24

The results presented in the table have been estimated via a TVP-VAR model with a first-order lag length that is based on the Akaike information criterion, a 20-step-ahead generalized forecast error variance decomposition (GFEVD), and a rolling window of 200 days. In the table, the "TO" column represents the connectedness effects transmitted by a market to other markets, whereas the "FROM" column indicates the connectedness effects received by a market from other markets. The "NET" column measures the net connectedness index, where markets with a positive NET value act as shock transmitters and those with a negative NET value act as shock receivers within the system. TCI denotes the total connectedness index. Each cell in the table reflects the connectedness from the market in the column to the market in the row

Panel B indicates that the conditional volatility connectedness is 48.78%. Volatility transmission within the cryptocurrency market ranges from 3.18% to 26.68%, highlighting a higher correlation within this market than others. However, the volatility connectedness between the cryptocurrency and stock markets is relatively small, ranging from 1.77% to 6.47%. Additionally, the volatility connectedness from the cryptocurrency market to the commodity and bond markets is low. The "FROM" connectedness index shows that DAX, Bitcoin, and Ethereum are the largest contributors, with values of 49.99%, 47.37%, and 47.96%, respectively, indicating their dominant role in volatility spillovers. According to the TO connectedness index, Ethereum, the S&P 500, and Bitcoin have the largest volatility connectedness effects transmitted to the entire system. Finally, the NET connectedness index reveals that the DAX and the SENSEX indices, USDT, and gold are net shock receivers, whereas the S&P 500 index, Bitcoin, Ethereum, the U.S. 5-year bond markets, and crude oil are net shock transmitters in terms of volatility.

Panel C shows that the total skewness connectedness is 38.12%, which is relatively lower than the total connectedness of the total volatility and kurtosis spillover. The diagonal elements reveal that Bitcoin (51.52%) and Ethereum (51.16%) have the lowest scores, whereas the highest scores are found for the S&P 500 index (77.86%), USDT (77.92%), crude oil (74.92%), and gold (73.49%). The diagonal elements for emerging and developed stock markets are notably higher than those of the cryptocurrency market. Concerning the TO connectedness index, Bitcoin (51.01%) and Ethereum (51.01%) have the strongest skewness spillovers to other markets, whereas the S&P 500 index (20.49%) and USDT (20.23%) have the weakest. This indicates limited international influence from the S&P 500 and USDT. In the "FROM" connectedness index, Bitcoin (48.48%) and Ethereum (48.84%) receive the most skewness spillovers from other markets. The S&P 500, Sensex indices, USDT, U.S. 5-year bonds, and gold are net receivers, whereas crude oil is the largest net transmitter, followed by Bitcoin, Ethereum, and the DAX index. Skewness indicates nuanced market responses to asymmetric information or unexpected crises, potentially intensifying spillovers to bad (negative skew) or good news (positive skew) compared with general price volatility.

Panel D shows the average connectedness results for kurtosis, revealing that the total spillover of kurtosis between cryptocurrency and other financial markets is 59.24%, which is notably greater than the total volatility spillover of 48.78% and the skewness spillover of 38.12%. This substantial spillover suggests that kurtosis exhibits greater synchronicity than volatility and skewness among cryptocurrencies and other financial markets. This indicates a tendency for market participants to respond uniformly to extreme events across these markets, possibly influenced by a common factor leading to high correlations in extreme volatility. The significant kurtosis spillover points to the susceptibility of the oil and stock markets to extreme events, resulting in sharper reactions to crises or unexpected economic news than normal fluctuations. Additionally, the elevated kurtosis spillover may illustrate a stronger transmission of tail risks between cryptocurrency and other financial markets. Among the assets assessed, the German stock market, Bitcoin, and Ethereum demonstrate the lowest diagonal elements, whereas the USDT, crude oil, and 5-year U.S. Treasury bond markets exhibit relatively higher kurtosis spillovers to the overall system. In contrast, the U.S. and Indian stock markets contribute the least to this connectedness. In terms of net connectedness, most markets, except for USDT, crude oil, and the 5-year U.S. Treasury bond markets, receive kurtosis spillovers, underscoring the interconnected nature of financial markets during extreme fluctuations.

Dynamic connectedness findings

To analyze the dynamic connectedness among cryptocurrencies, gold, crude oil, the 5-year U.S. bond, and stock markets, we estimate the dynamic total connectedness, net total directional connectedness, and net pairwise directional connectedness indices with a 200-day rolling window. We perform several robustness tests, starting with the estimation of the TVP-VAR model based on various assumptions regarding the selection of prior hyperparameters. The primary analysis of the TVP-VAR conducted here was estimated using a Bayesian VAR procedure, with a hyperparameter estimation size of 200 days. Given that this choice significantly influences the outcomes of the TVP-VAR model, we assessed whether the results obtained in the main empirical tests would be sensitive to different prior sizes. For robustness checks, we also select three window sizes. To assess the robustness of our findings relative to the selection of lookback periods, we re-estimate the TVP-VAR model using rolling windows of 150, 200, and 250 observations wide (Figs. 15, 16, and 17 in the Appendix) following the approach of Hanif et al. (2023). These windows were chosen on the basis of their mean absolute prediction error (MAPE) derived from a 20-step-ahead forecast. The TVP-VAR approach demonstrates greater robustness than the traditional rolling-window-based VAR, as it avoids the loss of significant observations, is less sensitive to outliers, and eliminates arbitrary decisions regarding the size of rolling windows (Antonakakis et al. 2020; Azimli 2024). This test ensures that the dynamic connectedness measures are insensitive to the choice of window length. The total connectedness index (TCI) results remain stable across all three specifications, exhibiting only marginal variation during peak spillover periods. This consistency reinforces the reliability of our model in capturing the underlying dynamics of market interactions.

Panels A-D in Fig. 3 show the dynamic total connectedness index (TCI) for return, volatility, skewness, and kurtosis over the markets, respectively. This figure captures the evolution of spillovers over time, showing an upward trend during periods of market turmoil.

Figure 3 shows that the total connectedness ranges between 20 and 70% for returns and 40% and 80% for volatility spillovers, reaching its highest levels at the beginning of 2018, 2020, and 2022. These dates correspond to significant events such as bubbles in the cryptocurrency market, the China–US trade war, the COVID-19 outbreak, and the Russia–Ukraine conflict. The fluctuations in the connectedness patterns reflect how markets respond to these financial and geopolitical crises. In 2018, the TCI exceeded 80% following the cryptocurrency bubble burst driven by manipulative news and the China–US trade war, which led to increased uncertainties and deterioration in investor sentiment in financial markets. Similarly, during the early stages of the COVID-19 outbreak in 2020, the total connectedness peaked at over 65% because of the widespread impact on major financial markets, including the cryptocurrency market. Although there was a gradual recovery during the outbreak, leading to a decline in the total connectedness index, the overall connectedness remained elevated due to ongoing global inflation uncertainty. In

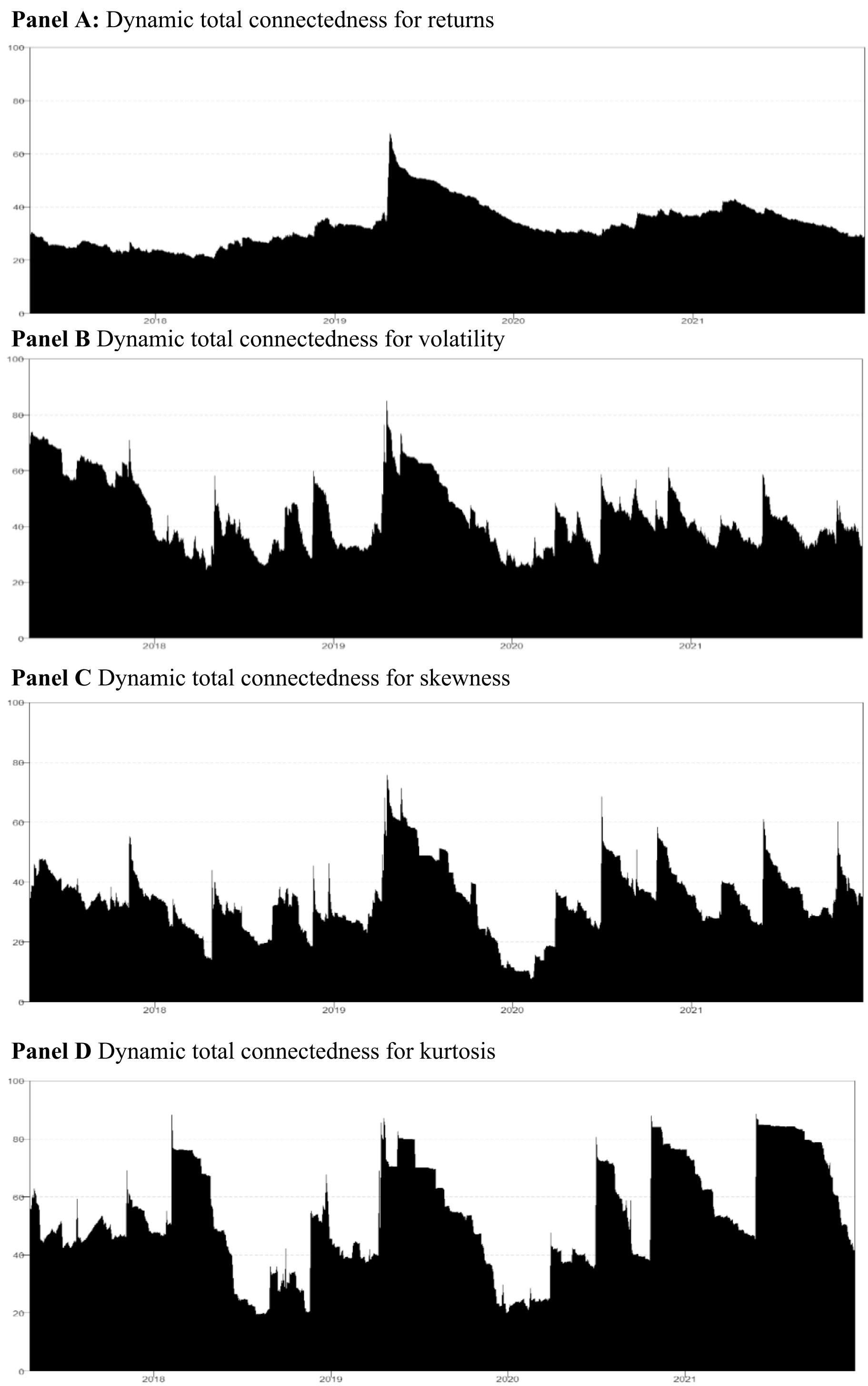


Fig. 3 Dynamic total connectedness for higher-order moments. *Note*: The results are based on a TVP-VAR model with a lag length of order one (BIC), a 20-step-ahead forecast, and a rolling window of 200 days, which corresponds to approximately one trading year. To ensure robustness, we also consider rolling window sizes of 150 and 250 days

2022, TCI reached 70% since the Russia–Ukraine conflict affected financial behaviors and skyrocketed uncertainty. The increase in total connectedness during these periods underscores the heightened interdependence among markets in times of stress. The dynamic total skewness connectedness changes between 10 and 80%.

The dynamic total skewness fluctuates between 10 and 80%, exhibiting several notable sharp increases throughout the examined period. The most significant increase, which occurred at the onset of the COVID-19 pandemic, increased from 40 to 80%. This surge indicates that the health crisis significantly amplified the total skewness connectedness. The pandemic profoundly affected oil supply and demand dynamics: lockdowns and reduced industrial activity diminished demand, whereas supply chain disruptions hindered production, resulting in considerable price volatility and increased skewness. Furthermore, the panic among investors during the pandemic led to drastic shifts in market sentiment, heightened market reactions and intensified skewness spillovers as investors reassessed risks and reallocated their assets. A subsequent sharp increase in skewness, from 79 to 93%, was observed after the outbreak of the Russia–Ukraine conflict in February 2022, highlighting the significant escalation of total skewness spillovers resulting from this geopolitical crisis. This observation is particularly noteworthy, as no previous studies have examined the skewness spillovers between the oil and stock markets during the Russia–Ukraine conflict.

With respect to the dynamic total kurtosis connectedness, notable time-varying characteristics are evident, with fluctuations ranging from 20 to 85%. The first major surge occurred in 2018, likely driven by the onset of the China–US trade war and the cryptocurrency bubble burst. Additionally, a sharp rise in dynamic total kurtosis connectedness was observed following the outbreak of the COVID-19 pandemic, underscoring its significant effect on extreme tail risk transmission among financial markets. Consequently, investors began to disregard their independent knowledge and increasingly imitated the investment strategies of others, leading to pronounced herding behaviors among cross-market investors. The Russia–Ukraine conflict has further reinforced kurtosis spillovers within the cryptocurrency–traditional financial system, as heightened investor panic, combined with the monetary policies of leading central banks, has exacerbated the situation.

The findings from the TCI indicate that the dynamic interconnections among the cryptocurrency, stock, 5-year U.S. Treasury bond, crude oil, and gold markets are significantly influenced by prevailing financial market conditions and the influence of economic and geopolitical risks. Notably, the dynamic total connectedness index tends to rise sharply during periods of extreme events. This phenomenon can be attributed to the substantial market shocks caused by these extreme occurrences, which result in dramatic declines in financial markets, increased economic uncertainty, and heightened market panic, ultimately leading to significant shifts in investor sentiment (Barunik and Křehlík 2018). It also highlights the importance of monitoring total dynamic connectedness during times of market stress to make informed investment decisions. These findings align with previous studies that have reported higher levels of market connectedness during periods of uncertainty (Zhang et al. 2023; Hanif et al. 2023).

Figure 4 shows the net directional connectedness for the cryptocurrency, gold, crude oil, 5-year US bond, and stock markets for higher-order moments. The net

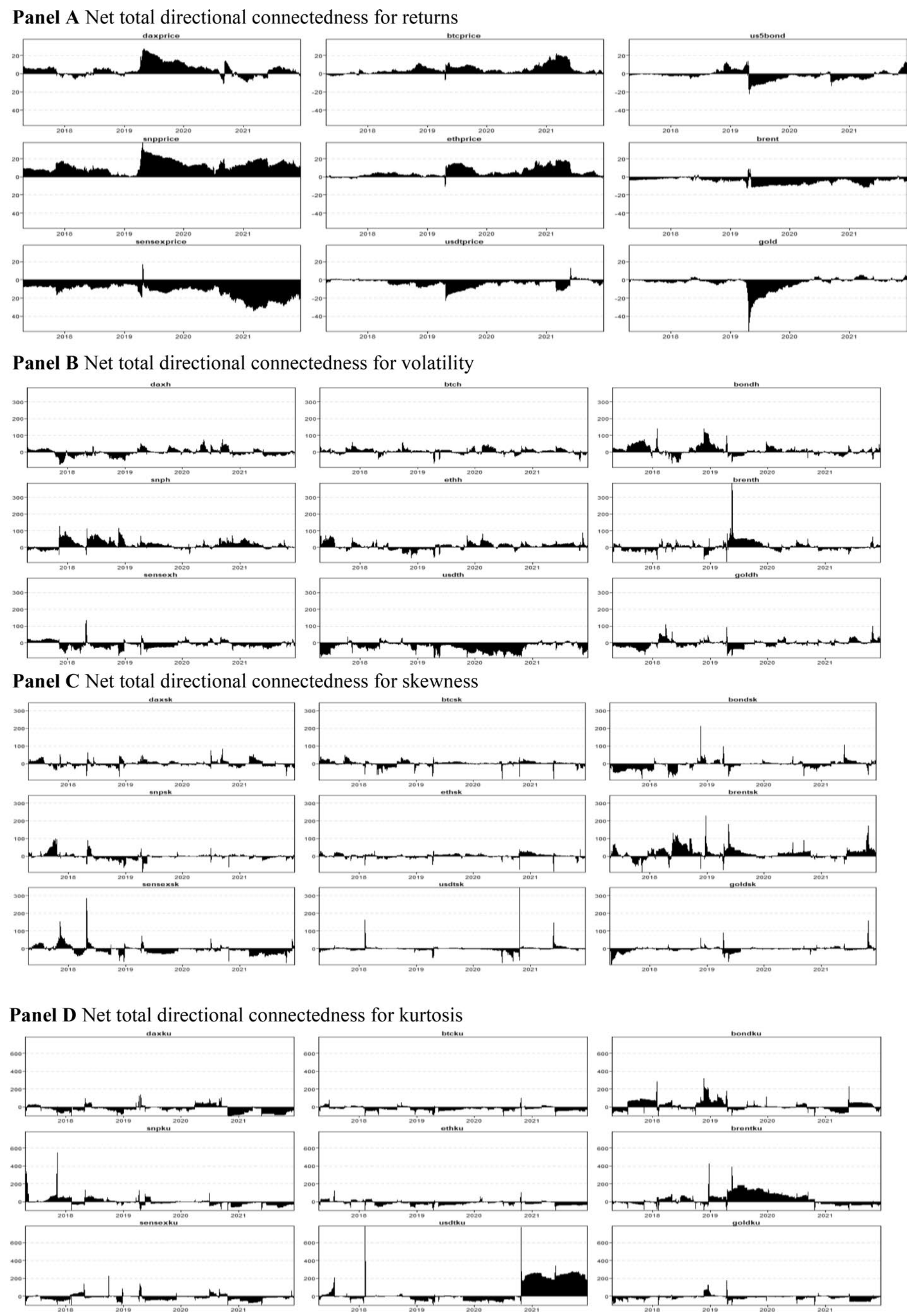


Fig. 4 Dynamics of the net total directional connectedness. index *Notes*: The results are based on a TVP-VAR model with a lag length of order one (BIC), a 20-step-ahead forecast, and a rolling window of 200 days, which corresponds to approximately one trading year. For robustness, we also consider rolling window sizes of 150 and 250 days

directional connectedness represents the variance between the "TO" and "FROM" spillovers. A positive net dynamic connectedness value signifies the market as a net transmitter of shocks to other markets, whereas a negative value indicates the market as a net receiver of shocks from other markets.

Panel A in Fig. 4 indicates that the net dynamic connectedness for returns varies over time. The SENSEX index, USDT, crude oil, and the 5-year U.S. bond market generally serve as net receivers of shocks from the entire system. Conversely, the DAX index, the S&P 500 index, Bitcoin, and Ethereum function function mainly as net transmitters of shocks for most of the sampling period. Gold usually acts as a shock receiver, except during financial and geopolitical crises. Interestingly, after 2020, gold transitioned to become a shock transmitter to other markets, particularly during the COVID-19 pandemic, the Russia–Ukraine conflict, and rising inflation. Additionally, the roles of Bitcoin and Ethereum as net shock transmitters became more evident during extreme events, suggesting that investors are increasingly considering cryptocurrencies alongside gold as safe-haven assets.

Panel B illustrates the net dynamic connectedness for volatility, showing that Bitcoin consistently transmits net volatility shocks most of the time. Ethereum initially acted as a net receiver until early 2020 but changed to being a net transmitter after mid-2021, coinciding with ongoing economic and geopolitical risks such as the COVID-19 pandemic and the Russia–Ukraine war. The USDT maintains a net receiver role throughout the sample. The DAX index received volatility spillovers notably in 2018, 2019, and after 2022 but transmitted significant net skewness spillovers afterward. The S&P 500 index consistently transmits kurtosis spillovers, whereas the SENSEX index remains a net receiver. Both the 5-year U.S. bond market and crude oil operate as both recipients and transmitters of volatility spillovers, yet gold continues to be a net shock receiver.

In Panel C, the dynamic net spillovers of skewness are examined. Here, Bitcoin, Ethereum, and USDT consistently act as net recipients of skewness spillovers across the entire sample. The U.S., German, and Indian stock markets also function as net receivers. The crude oil market and the Russian stock market transmitted the highest net skewness spillovers in 2018, 2019, and early 2022 but subsequently started receiving considerable net skewness spillovers. Furthermore, both the 5-year U.S. Treasury bond and gold are recipients of net skewness.

Panel D depicts the dynamic net spillovers of kurtosis, showing fluctuations over time. The U.S., German, and Indian stock markets generally remain consistent net receivers of kurtosis spillovers throughout the sample. Bitcoin and Ethereum also receive kurtosis spillovers. In contrast, USDT becomes the main source of kurtosis spillovers, influencing other markets more than being influenced itself after 2021. Crude oil transmitted more net kurtosis spillovers in 2018 and early 2022 than it received. The 5-year U.S. Treasury bond transmitted net kurtosis spillovers during 2018, 2019, and 2022, corresponding with events such as the China–U.S. trade war, the COVID-19 pandemic, and the Russia–Ukraine war, whereas gold consistently remained a net receiver of kurtosis spillover throughout the period analyzed.

Importantly, while the net total connectedness results are useful for identifying net transmitters and receivers within a network, they provide only a partial understanding of the underlying dynamics. These results do not fully capture the pairwise interactions between variables over time. To gain a more nuanced understanding of the specific roles each variable plays within the network and how these roles evolve, it is essential to analyze the bilateral outcomes. This approach offers valuable insights into the linkages between cryptocurrency and financial markets, enriching the discussion around their

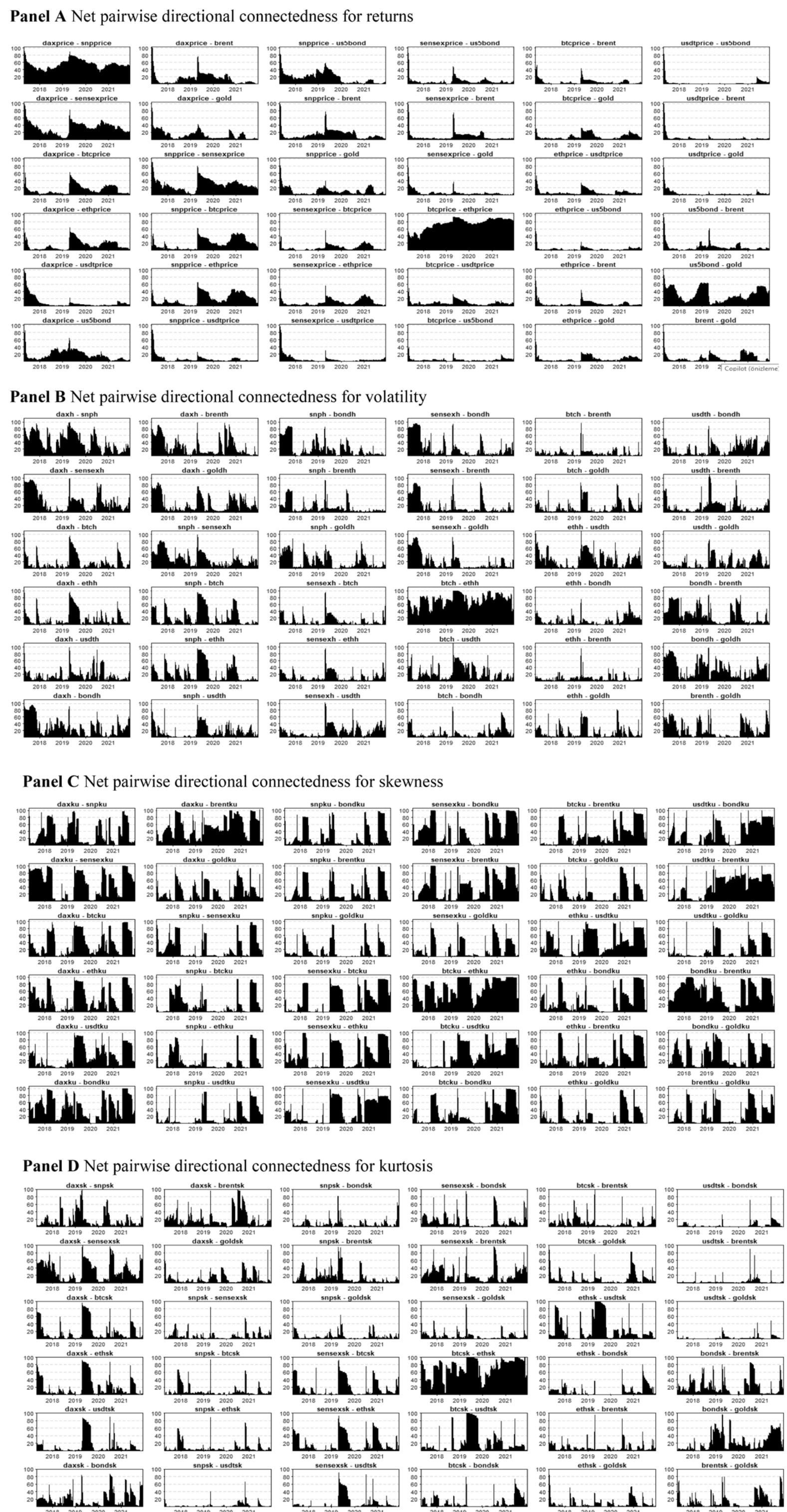


Fig. 5 Net pairwise directional connectedness. *Notes*: The results are based on a TVP-VAR model with a lag length of order one (BIC), a 20-step-ahead forecast and a rolling window of 200 days. The rolling window of 200 days is for a trading year. We also consider window sizes of 150 and 250 days to obtain robust results

interconnectivity. Examining pairwise dynamics is therefore crucial for achieving a comprehensive understanding of the relationships between variables in a network.

Figure 5 presents the net pairwise directional connectedness for returns, volatility, skewness, and kurtosis spillovers. In Panel A, the pairwise connectedness results for returns indicate that the DAX, S&P 500, and SENSEX indices consistently act as key transmitters of shocks to all three cryptocurrencies, evident through positive pairwise connectedness indices throughout the observed period. Moreover, the findings show that the magnitude of shock transmission between the cryptocurrency market and traditional financial markets is nonlinear and varies over time. This suggests that the cryptocurrency market is becoming increasingly integrated with the traditional financial system and is establishing itself as a significant player in global financial markets. These conclusions are crucial for investors and policymakers, emphasizing the need to comprehend the evolving relationship between cryptocurrency markets and financial systems to make informed decisions.

In Panel B, the volatility results indicate that stock markets act as volatility spillover transmitters to the cryptocurrency market. Additionally, the cryptocurrency market emerges as a source of volatility spillovers to the 5-year U.S. bond and commodity markets, reflected in positive values. The degree of volatility transmission between these markets fluctuates over time, particularly during extreme events. This finding aligns with studies by Karau (2023) and Pietrzak (2023), which suggest that while cryptocurrency prices were not previously influenced by U.S. monetary policy, they began to respond similarly to other risky assets in late 2020. Historical data show that U.S. monetary tightening has often led to an increase in Bitcoin prices, highlighting Bitcoin's speculative nature and its systematic response to monetary and central bank information shocks.

Panel C discusses the dynamic net pairwise skewness spillovers, revealing that cryptocurrencies transmit spillovers to the gold, bonds, crude oil, and stock markets. Notably, the cryptocurrency market displays strong and consistent net skewness spillovers to the German and Indian stock markets, particularly during 2020 and 2021. Conversely, the U.S. stock market, the 5-year Treasury bond, and gold receive comparatively less net skewness spillover from cryptocurrencies.

As demonstrated in Panel D, dynamic net pairwise kurtosis spillovers are largely positive, suggesting that cryptocurrencies, bonds, crude oil, and gold transmit significant net kurtosis spillovers to stock markets. Specifically, these markets transmit greater net kurtosis spillovers to the German and Indian stock markets during 2018, 2020, 2021, and early 2022, whereas the S&P 500 experiences the lowest net kurtosis spillovers during the same periods. Moreover, the interactions between the cryptocurrency/gold markets and the U.S. stock market in terms of kurtosis spillovers are minimal, indicating a lack of significant influence or interaction.

Conclusively, the analysis of net total directional and pairwise indices offers crucial insights into which cryptocurrencies effectively transmit market shocks, guiding portfolio diversification strategies. By combining shock transmitters with shock-resilient cryptocurrencies, investors can significantly enhance their risk profiles. Diversification opportunities shift during financial crises, making it essential for investors to focus on holding shock transmitters while minimizing exposure to shock receivers, as transmitters are affected by fewer risk factors. Furthermore, our study reveals cryptocurrencies' potential as safe havens compared with traditional assets such as gold, bonds, and stocks. During market stress, cryptocurrencies often maintain or appreciate, owing to their intrinsic scarcity and commodity-like nature, which protects them from currency fluctuations. By including these assets in their portfolios, investors can mitigate risks and potentially achieve better returns (Bouri et al. 2021a, b, c, d; Azimli 2024).

Empirical ınsights into higher-order moment spillovers

Skewness and kurtosis are essential in analyzing return distributions. Skewness indicates the asymmetry of returns across markets, whereas kurtosis measures the tail thickness and the likelihood of extreme events (Xu et al. 2024). A positively skewed distribution shows frequent small losses with few notable gains, whereas a negatively skewed distribution reflects many small gains and rare significant losses. Analyzing skewness spillovers reveals how markets interconnect with respect to return asymmetry, impacting both downside and upside risk. This insight highlights the informational efficiency of financial markets in processing cross-border information. Additionally, higher kurtosis signifies increased investment risk due to a greater probability of extreme returns (He and Hamori 2021). Understanding these factors is critical for informed investment strategies.

The connectedness results for higher-order moments, particularly skewness and kurtosis, provide valuable insights into the asymmetric transmission of risk across financial markets. The total connectedness of skewness across markets is lower than that of volatility and kurtosis. Bitcoin and Ethereum are both strong transmitters and receivers of skewness shocks, indicating that they play a central role in the asymmetric distribution of returns. Notably, skewness connectedness is greater during major events, such as the COVID-19 pandemic and the collapse of the FTX, revealing heightened sensitivity to negative tail risk during crises.

Kurtosis connectedness indicates greater synchronization of extreme events across markets. Crude oil, USDT, and the U.S. 5-year bond appear to be the dominant transmitters of extreme tail risk. In contrast, stock indices such as the S&P 500 and Sensex act more as receivers, suggesting a delayed response to systemic shocks originating in other markets. These results emphasize the importance of monitoring fourth-moment dynamics, especially under turbulent conditions where extreme deviations from the mean are more frequent.

The dynamic total skewness significantly increased at the onset of the COVID-19 pandemic. Lockdowns curtailed demand, whereas supply chain disruptions hindered production, resulting in price volatility and heightened skewness. Investor panic during this time led to dramatic shifts in market sentiment, intensifying skewness spillovers as asset reallocations occurred. A further surge in skewness was recorded after the outbreak of the Russia–Ukraine conflict in February 2022, indicating a significant escalation tied to this geopolitical crisis. With respect to dynamic total kurtosis connectedness, the first major surge occurred in 2018, driven by the China–U.S. trade war and the collapse of the cryptocurrency bubble. Another notable rise was observed following the COVID-19 pandemic, impacting the transmission of extreme tail risk among financial markets. Investors increasingly adopt the strategies of others, resulting in pronounced herding behavior. The Russia–Ukraine conflict has further exacerbated kurtosis spillovers between cryptocurrencies and traditional financial systems, fueled by heightened investor panic and the monetary policies of central banks.

Together, these findings highlight the limitations of traditional mean–variance approaches, which may underestimate tail risk exposure. Investors and portfolio managers could benefit from integrating higher-moment considerations into their allocation strategies. For example, allocating away from assets with high positive kurtosis during stressed periods could reduce exposure to extreme downside shocks. Similarly, skewness-aware optimization techniques may improve risk-adjusted returns, especially when return distributions are nonnormal.

Hedging ratio and minimum connectedness portfolio strategy

To uncover the minimum connectedness portfolio strategy, we delve into the interplay between cryptocurrency, commodity, bond, and stock markets by analyzing hedging ratios and implementing minimum connectedness portfolio strategies. These analyses are crucial for understanding how investors can optimize their portfolios in a highly interconnected financial environment. By examining asset weights, hedge effectiveness, and portfolio construction techniques, we aim to provide insights into the effectiveness of various hedging strategies. This not only helps investors manage risk more effectively but also highlights the role of cryptocurrencies in modern portfolio management. The findings discussed here are integral to forming strategies that can enhance portfolio performance, particularly during periods of market volatility and economic uncertainty. Table 6 presents the results for average asset weights in bivariate portfolios and hedge effectiveness (HE), which refers to the percentage reduction in the variance of an unhedged position. The "Mean" and "Std. Dev." columns report the average weights and standard deviations of the diversification assets. A higher HE value indicates greater risk reduction. By leveraging the average values shown in Table 6, investors can construct portfolios with confidence and aim to maximize their returns. The results show that Bitcoin and Ethereum hold the highest asset weights in the return portfolio from April 2017 to December 2023, a period that includes the COVID-19 pandemic, the Russia–Ukraine war, and the Silicon Valley Bank crisis. For instance, the mean value of the Bitcoin/gold (btcprice/gold) ratio is 0.42, indicating that Bitcoin accounts for 42% of the portfolio's weight. This implies that a hedge ratio of 0.42 means that for every $1 invested in Bitcoin, 42 cents should be allocated to gold. Additionally, asset weights for cryptocurrencies are significantly higher than those for gold and crude oil. The HE column in Panel A of Table 6 indicates that the stock market provides a higher level of hedge effectiveness (ranging from 2 to 38%) in the cryptocurrency market. Similarly, cryptocurrencies demonstrate greater hedge effectiveness (ranging from 4 to 10%) than gold (2% to 5%) and crude oil (3% to 9%) do on the stock market. Among the asset pairs, the gold/US 5-year bond (gold/us5bond) has the highest hedge efficiency, whereas the Bitcoin/US 5-year bond (btcprice/us5bond) and Ethereum/US 5-year bond (ethprice/us5bond) have the lowest hedge efficiency.

While the hedge ratio is a useful metric for hedging, it can be influenced by outliers and does not provide insights into portfolio weights. To address this limitation, Broadstock et al. (2022) propose a minimum portfolio weight approach that emphasizes the interconnectedness of variables and is more robust to network shocks. In our research,

Table 6 Hedging effectiveness results

	Mean	Std.Dev	HE	*p* value		Mean	Std.Dev	HE	*p* value		Mean	Std.Dev	HE	*p* value
Panel A. Return portfolio														
daxprice/snpprice	0.60	0.12	0.36	0.20	btcprice/daxprice	0.64	0.50	0.07	0.41	us5bond/daxprice	0.80	0.52	0.11	0.41
daxprice/sensexprice	0.45	0.15	0.22	0.64	btcprice/snpprice	0.93	0.6	0.10	0.20	us5bond/snpprice	0.63	0.63	0.14	0.20
daxprice/btcprice	0.06	0.06	0.11	0.53	btcprice/sensexprice	0.31	0.50	0.04	0.64	us5bond/brent	0.17	0.13	0.05	0.40
daxprice/ethprice	0.05	0.05	0.11	0.57	btcprice/ethprice	0.59	0.12	0.59	0.57	us5bond/gold	−1.41	0.53	0.13	0.00
daxprice/usdtprice	−1.07	1.64	0.02	0.85	btcprice/usdtprice	1.18	9.27	0.04	0.85	brent/daxprice	0.48	0.48	0.09	0.41
daxprice/us5bond	0.09	0.07	0.10	0.00	btcprice/us5bond	0.00	0.16	0.01	0.00	brent/snpprice	0.54	0.42	0.09	0.20
daxprice/brent	0.07	0.07	0.06	0.40	btcprice/brent	0.09	0.12	0.01	0.40	brent/sensexprice	0.31	0.37	0.03	0.64
daxprice/gold	−0.03	0.28	0.04	0.00	btcprice/gold	0.42	0.77	0.03	0.00	brent/btcprice	0.07	0.11	0.03	0.53
snpprice/daxprice	0.56	0.11	0.38	0.41	ethprice/daxprice	0.95	0.51	0.07	0.41	brent/ethprice	0.06	0.08	0.03	0.57
snpprice/sensexprice	0.33	0.14	0.13	0.64	ethprice/snpprice	1.42	0.60	0.10	0.20	brent/usdtprice	−0.69	2.56	0.01	0.85
snpprice/btcprice	0.08	0.07	0.17	0.53	ethprice/sensexprice	0.54	0.80	0.04	0.64	brent/us5bond	0.12	0.07	0.05	0.00
snpprice/ethprice	0.06	0.05	0.18	0.57	ethprice/btcprice	1.02	0.16	0.58	0.53	brent/gold	0.47	0.66	0.04	0.00
snpprice/usdtprice	0.47	2.51	0.02	0.85	ethprice/usdtprice	1.68	12.26	0.03	0.85	gold/daxprice	−0.03	0.12	0.03	0.41
snpprice/us5bond	0.07	0.09	0.14	0.00	ethprice/us5bond	0.01	0.17	0.01	0.00	gold/snpprice	0.03	0.13	0.05	0.20
snpprice/brent	0.08	0.05	0.09	0.40	ethprice/brent	0.18	0.13	0.02	0.40	gold/sensexprice	−0.01	0.07	0.02	0.64
snpprice/gold	0.08	0.23	0.06	0.00	ethprice/gold	0.76	0.61	0.03	0.00	gold/btcprice	0.02	0.03	0.04	0.53
sensexprice/daxprice	0.31	0.11	0.24	0.41	usdtprice/daxprice	−0.03	0.06	0.01	0.41	gold/ethprice	0.02	0.02	0.04	0.57
sensexprice/snpprice	0.25	0.14	0.13	0.20	usdtprice/snpprice	0.00	0.04	0.01	0.20	gold/usdtprice	0.68	1.11	0.01	0.85
sensexprice/btcprice	0.02	0.04	0.05	0.53	usdtprice/sensexprice	−0.02	0.04	0.01	0.64	gold/us5bond	−0.11	0.06	0.19	0.00
sensexprice/ethprice	0.02	0.03	0.06	0.57	usdtprice/btcprice	0.00	0.01	0.03	0.53	gold/brent	0.03	0.04	0.04	0.40
sensexprice/usdtprice	−0.36	1.09	0.02	0.85	usdtprice/ethprice	0.00	0.01	0.02	0.57					
sensexprice/us5bond	0.04	0.02	0.05	0.00	usdtprice/us5bond	0.00	0.01	0.00	0.00					
sensexprice/brent	0.03	0.04	0.04	0.40	usdtprice/brent	0.00	0.01	0.00	0.40					
sensexprice/gold	−0.01	0.10	0.02	0.00	usdtprice/gold	0.03	0.05	0.01	0.00					

Mean and Std. Dev. columns represent the average weights and standard deviation of the hedge ratio of diversification assets. The HE column shows the hedge effectiveness we construct minimum variance, minimum correlation, and minimum connectedness portfolios via the TVP-VAR model for the cryptocurrency, commodity, bond, and stock markets. This approach helps us determine the minimum portfolio weights required to achieve optimal outcomes. In portfolio and risk management, diversification is crucial, and selecting the appropriate portfolio construction technique is key to achieving this goal. Therefore, we examine the implications of using the minimum variance portfolio (MVP), minimum correlation portfolio (MCP), and minimum connectedness portfolio (MCoP) approaches by comparing their hedging effectiveness scores. Investors who prioritize risk management over returns typically opt for a minimum volatility portfolio (MVP), which includes assets characterized by low volatility to help minimize risk and stabilize portfolio returns during turbulent market conditions. Those who favor diversification along with risk management may select a minimum correlation portfolio (MCP) by incorporating assets that exhibit low correlation with one another. Additionally, investors concerned about macroeconomic risk may choose a macroeconomic coherence portfolio (MCoP). These investors focus on assets that are not significantly affected by the same economic and financial factors, aiming to reduce risk and provide a safeguard against macroeconomic shocks that can impact various asset classes (Rubbaniy et al. 2024).

Table 7 presents the descriptive statistics of the portfolio weights and hedge effectiveness. Overall, our portfolio analysis supports the presence of a dynamic network that allows for diversification opportunities. For the MVP, if an investor allocates 4% to the DAX index, 2% to the S&P 500 index, 7% to the SENSEX index, 0% to Bitcoin, 0% to Ethereum, 72% to USDT, 2% to the 5-year U.S. bond, 0% to crude oil, and 13% to gold, the impact of each asset's volatility on the portfolio would be reduced by 97% from the DAX index, 97% from the S&P 500 index, 97% from the SENSEX index, 100% from Bitcoin, 100% from Ethereum, 73% from USDT, 100% from the 5-year U.S. bond, 100% from crude oil, and 94% from gold. However, according to the MCP, if an investor allocates 9% to Bitcoin, 5% to Ethereum, 21% to the 5-year US bond, and 8% to crude oil, the volatility impact would be reduced by 93% from Bitcoin, 96% from Ethereum, 91% from the 5-year US bond, and 87% from crude oil. Similarly, under the MCoP approach, if an investor allocates 9% to Bitcoin, 8% to Ethereum, 12% to the 5-year US bond, and 15% to crude oil, the volatility impact would be reduced by 91% from Bitcoin, 95% from Ethereum, 90% from the 5-year US bond, and 84% from crude oil.

The cumulative returns of the portfolios constructed via three different methods are displayed in Fig. 6, specifically analyzing the MVP, MCP, and MCoP. Notably, there was a sharp decline in cumulative returns in 2020 during the initial wave of COVID-19, followed by a gradual recovery until early 2022. Returns peaked again in the wake of the Russia–Ukraine conflict, with the MCP and MCoP maintaining positive returns, whereas the MVP's returns lagged behind those of the other two portfolios during both the post-COVID-19 phase and the Russia–Ukraine crisis. Furthermore, the cumulative returns associated with the connectedness portfolios based on volatility, skewness, and kurtosis significantly decreased. Specifically, prior to the COVID-19 pandemic, the skewness connectedness portfolio had relatively lower cumulative returns than the other connected portfolios did. Interestingly, the cumulative returns of kurtosis connectedness peaked during the COVID-19 pandemic but decreased again afterwards, although they

Table 7 Multivariate portfolio descriptive statistics for diversification strategies

	Mean	Std.Dev	HE	*p* value	Mean	Std.Dev	HE	*p* value
Minimum correlation portfolios (MCP)								
	Panel A. Return				Panel B. Volatility			
Daxprice	0.05	0.04	−0.03	0.5200	0.05	0.04	−167.88	0.0000
Snpprice	0.02	0.02	−0.03	0.6100	0.06	0.06	−6.83	0.0000
Sensexprice	0.11	0.03	−0.25	0.0000	0.12	0.04	−104.92	0.0000
Btcprice	0.09	0.03	0.93	0.0000	0.07	0.04	−12.08	0.0000
Ethprice	0.05	0.04	0.96	0.0000	0.09	0.04	0.89	0.0000
Usdtprice	0.17	0.04	−8.91	0.0000	0.18	0.02	−710.82	0.0000
us5bond	0.21	0.05	0.91	0.0000	0.14	0.03	0.32	0.0000
Brent	0.08	0.02	0.87	0.0000	0.17	0.04	0.03	0.5400
Gold	0.22	0.04	−1.09	0.0000	0.11	0.03	−313.7	0.0000
	Panel C. Skewness				Panel D. Kurtosis			
Daxprice	0.05	0.04	−167.88	0.0000	0.03	0.04	−21.52	0.0000
Snpprice	0.06	0.06	−6.83	0.0000	0.15	0.03	0.98	0.0000
Sensexprice	0.12	0.04	−104.92	0.0000	0.1	0.04	−17.92	0.0000
Btcprice	0.07	0.04	−12.08	0.0000	0.1	0.13	−0.01	0.8800
Ethprice	0.09	0.04	0.89	0.0000	0.08	0.07	−23.6	0.0000
Usdtprice	0.18	0.02	−710.82	0.0000	0.15	0.02	0.49	0.0000
us5bond	0.14	0.03	0.32	0.0000	0.13	0.03	−14.21	0.0000
Brent	0.17	0.04	0.03	0.54	0.13	0.05	0.3	0.0000
Gold	0.11	0.03	−313.7	0.0000	0.13	0.04	−1212.05	0.0000
Minimum variance portfolios (MVP)								
	Panel A. Return				Panel B. Volatility			
Daxprice	0.04	0.04	0.97	0.0000	0.15	0.17	0.62	0.0000
Snpprice	0.02	0.04	0.97	0.0000	0	0	0.98	0.0000
Sensexprice	0.07	0.08	0.97	0.0000	0.09	0.12	0.76	0.0000
Btcprice	0	0	1	0.0000	0	0	0.97	0.0000
Ethprice	0	0	1	0.0000	0	0	1	0.0000
Usdtprice	0.72	0.32	0.73	0.0000	0.64	0.41	−0.59	0.0000
us5bond	0.02	0.02	1	0.0000	0	0	1	0.0000
Brent	0	0	1	0.0000	0	0	1	0.0000
Gold	0.13	0.15	0.94	0.0000	0.11	0.14	0.3	0.0000
	Panel C. Skewness				Panel D. Kurtosis			
Daxprice	0.15	0.17	0.62	0.0000	0.1	0.09	0.66	0.0000
Snpprice	0	0	0.98	0.0000	0	0	1	0.0000
Sensexprice	0.09	0.12	0.76	0.0000	0.05	0.05	0.71	0.0000
Btcprice	0	0	0.97	0.0000	0	0	0.98	0.0000
Ethprice	0	0	1	0.0000	0.36	0.25	0.62	0.0000
Usdtprice	0.64	0.41	−0.59	0.0000	0.01	0.01	0.99	0.0000
us5bond	0	0	1	0.0000	0.03	0.06	0.77	0.0000
Brent	0	0	1	0.0000	0	0	0.99	0.0000
Gold	0.11	0.14	0.3	0.0000	0.46	0.23	−17.55	0.0000
Minimum connectedness portfolios (MCoP)								
	Panel A. Return				Panel B. Volatility			
Daxprice	0.06	0.03	−0.23	0.0000	0.07	0.05	−71.64	0.0000
Snpprice	0.06	0.03	−0.22	0.0000	0.07	0.05	−2.37	0.0000
Sensexprice	0.13	0.02	−0.5	0.0000	0.13	0.05	−44.56	0.0000
Btcprice	0.09	0.03	0.91	0.0000	0.11	0.08	−4.63	0.0000
Ethprice	0.08	0.03	0.95	0.0000	0.1	0.06	0.95	0.0000

Table 7 (continued)

	Mean	Std.Dev	HE	*p* value	Mean	Std.Dev	HE	*p* value
Usdtprice	0.17	0.02	− 10.83	0.0000	0.12	0.06	− 305.19	0.0000
us5bond	0.12	0.04	0.9	0.0000	0.12	0.06	0.71	0.0000
Brent	0.15	0.02	0.84	0.0000	0.16	0.04	0.58	0.0000
Gold	0.13	0.02	− 1.49	0.0000	0.12	0.05	− 134.37	0.0000
	Panel C. Skewness				Panel D. Kurtosis			
Daxprice	0.07	0.05	− 71.64	0.0000	0.04	0.05	− 37.49	0.0000
Snpprice	0.07	0.05	− 2.37	0.0000	0.17	0.03	0.97	0.0000
Sensexprice	0.13	0.05	− 44.56	0.0000	0.11	0.03	− 31.33	0.0000
Btcprice	0.11	0.08	− 4.63	0.0000	0.07	0.09	− 0.72	0.0000
Ethprice	0.1	0.06	0.95	0.0000	0.09	0.07	− 41.03	0.0000
Usdtprice	0.12	0.06	− 305.19	0.0000	0.14	0.03	0.12	0.0000
us5bond	0.12	0.06	0.71	0.0000	0.1	0.06	− 24.99	0.0000
Brent	0.16	0.04	0.58	0.0000	0.14	0.04	− 0.19	0.0000
Gold	0.12	0.05	− 134.37	0.0000	0.15	0.03	− 2071.81	0.0000

This table presents the average and standard deviation of portfolio weights, along with the hedging effectiveness (HE) for each asset

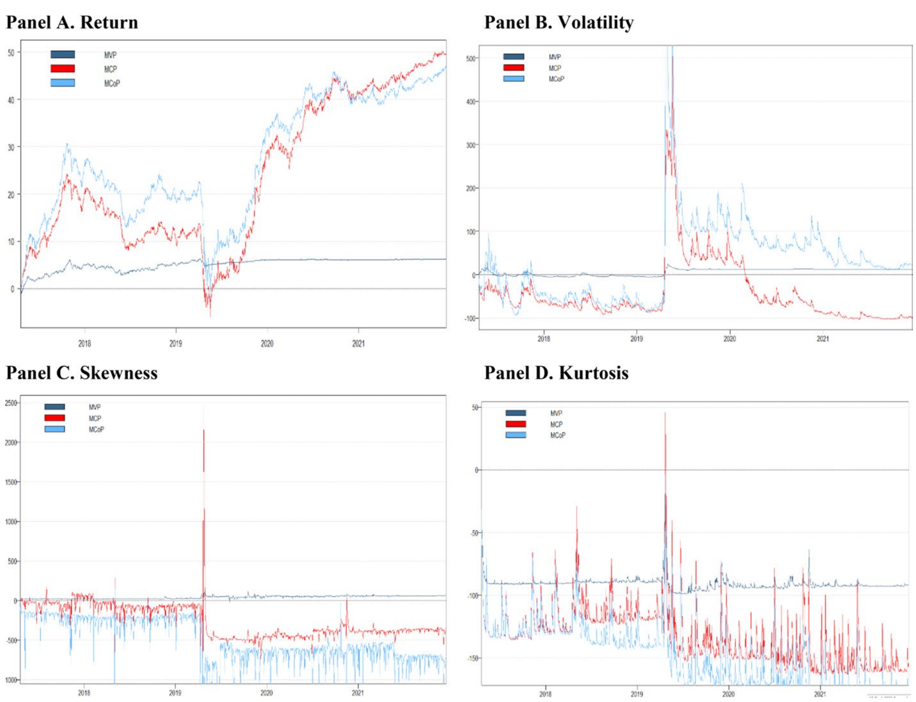


Fig. 6 Cumualtive portfolio returns for the higher-order moments

remained positive in the MCP and MCoP. Figure 6 shows that the minimum kurtosis connectedness portfolio had the highest cumulative returns, followed by those of the minimum skewness and minimum volatility connectedness portfolios.

In summary, the analysis indicates that during the peak period of COVID-19, the MCoP significantly outperforms the other two portfolios, highlighting its potential to shield investors from market volatility and systemic risk during health crises. This performance aligns with findings from Abdullah et al. (2023), Cui and Maghyereh (2023), and Rubbaniy et al. (2024), which suggest that during both the COVID-19 pandemic and the Russia–Ukraine conflict, investors, particularly hedgers and those averse to risk, favor minimum connectedness strategies to mitigate risks. Our results advocate considering asset interconnectedness as a strategy for building more resilient portfolios, especially in times of crisis. The MCP also showed commendable performance, whereas the MVP indicated limitations under extreme market conditions, suggesting that a focus beyond volatility may enhance performance. Moreover, the findings highlight the benefits of higher-order moment connectedness in improving multivariate portfolio returns, providing valuable insights for portfolio managers aiming to optimize risk management and return strategies.

Portfolio rebalancing involves adjusting an investment portfolio to maintain a desired asset allocation over time. This process aims to minimize risk and maximize returns by responding to fluctuations in asset prices, correlations, and prevailing market conditions. The TCI, which is a tool that helps investors determine how various assets within a portfolio are interconnected, allows them to assess the correlation between assets in their portfolio. This insight enables investors to evaluate whether their portfolio is genuinely diversified or if it is overly exposed to risk factors that could cause all assets to move in tandem or exhibit high correlations. A high TCI indicates that assets are closely correlated, which increases overall risk, whereas a low TCI suggests better diversification. Improving diversification and decreasing reliance on highly connected assets to mitigate this exposure may be wise. As the TCI fluctuates with market conditions, portfolios should be rebalanced periodically (e.g., monthly or quarterly) to optimize the risk-return profile (Diebold and Yilmaz 2012, 2014).

Maintaining an optimized portfolio requires continuous rebalancing to account for market fluctuations and ensure alignment with risk and return objectives. Rebalancing strategies can be time-based or triggered by significant deviations from target weights, helping to avoid overconcentration in specific assets. Financial crises such as the 2008 global financial crisis and the COVID-19 pandemic present challenges for portfolio management, as they disrupt markets and change asset correlations, which reduces traditional diversification. Therefore, dynamic rebalancing strategies are essential. For example, with a TCI below 20% (low systemic risk), investors should increase exposure to risky assets such as Bitcoin and Ethereum. During a normal TCI range of 20–50%, a neutral allocation is recommended. However, if the TCI exceeds 0.50% (high systemic risk), investors should shift their weight from risky assets to safe havens such as gold and Treasury bonds. In turbulent periods, such as during the COVID-19 pandemic, when TCI is above 50%, reallocating from riskier assets to safer options is advisable. This approach maintains defensive positions during volatility while allowing for aggressiveness during calmer periods. TVP-VAR-informed dynamic portfolio reallocation can enhance risk-adjusted returns and reduce volatility, leading to lower drawdowns and better capital preservation during high-TCI periods. This model effectively captures market shifts, resulting in improved cumulative performance (as shown in Fig. 6) over time (Fahlenbrach et al. 2021; Cui and Maghyereh 2023; Yousaf et al. 2024).

Prior sensitivity analysis

We conduct a sensitivity analysis to assess the impact of different prior assumptions on the results. For this analysis, we employ one uninformative prior and 500 random Minnesota priors, following the approach of Antonakakis et al. (2020). The uninformative prior is defined as follows:

$$vec(A_0) \sim N\left(vec(0), m^{-p}.I\right)$$
$$\sum_0 = Cov\left(y\right)$$

The Minnesota prior is calculated as follows:

$$vec(A_0) \sim N\left(vec\left(diag(U(m, -0.99, 0.99))\right), m^{-p}.I\right)$$
$$\sum_0 = Cov\left(y\right)$$

In this approach, the coefficients with a one-lag order are assigned a random and uniform distribution, whereas the other coefficients are set to zero. This prior assumes that all assets follow a univariate process, which can vary in persistence from highly persistent to a random walk process. The parameters are constrained within a range of −0.99 to 0.99, ensuring that no series is nonstationary. By randomly selecting these parameters, we can explore different settings and observe the resulting variations in our analysis. Our findings, which are based on 500 randomly chosen Minnesota priors following the methodology of Antonakakis et al. (2020), show that the results are consistent across all priors. As illustrated in Figs. 9, 10, and 11 in the Appendix, the outcomes remain stable even after limited updating of the coefficients.

Forgetting factor sensitivity analysis

Following the approach of Antonakakis (2020), we perform a forgetting factor sensitivity analysis to assess the stability of the results for total dynamic connectedness, net total connectedness, and pairwise directional connectedness. Koop and Korobilis (2013) suggested that while the forgetting factors can change over an additional process, this may introduce a computational burden without necessarily increasing the forecasting performance. To address this, we let two forgetting factors (κ1, κ2) change irrespective of each other between 0.96 and 0.99.

Following Antonakakis et al. (2020) and Bouri et al. (2021a, b, c, d), we use a Kalman filter estimation with forgetting factors that allow variance to change over time, setting the decay factors to 0.99 for the VAR forgetting factor and 0.96 for the exponentially weighted moving average (EWMA) forgetting factor. These choices undergo robustness tests for outlier insensitivity. The Kalman filter is a key tool for estimating the state of dynamic systems from noisy data, serving as the optimal linear estimator for linear Gaussian systems. Its performance, however, declines with outliers. To enhance robustness, research has explored nonGaussian distributions, as multivariate Gaussian models are sensitive to outliers. For example, some studies utilize multivariate Student-t distributions, although these can complicate parameter estimation. Alternative approaches include modeling noise as nonGaussian distributions or employing resampling techniques and numerical integration, although these often require substantial computation.

Additionally, a weighted least squares method assigns statistical properties to measurement residuals, enhancing the Kalman filter's robustness to outliers by weighting data samples on the basis of their contribution to the hidden state estimate (Morris 1976; Huber 1964; Sorenson and Alspach 1971; Masreliez 1975; Kramer and Sorenson 1988).

Estimating the connectedness measures via the TVP-VAR model with these varying forgetting factors provides additional insights into the stability of our results. Figures 12, 13, and 14 in the Appendix illustrate the sensitivity of total dynamic connectedness, net total connectedness, and pairwise directional connectedness to the independently varying forgetting factors. These findings further underscore the robustness and significance of our results.

Model capacity to capture structural shifts

Rather than imposing arbitrary structural breakpoints, the TVP-VAR framework used in this study is specifically designed to endogenously capture regime shifts and time-varying dynamics. Its state-space representation with stochastic volatility allows for smooth or abrupt transitions in relationships across markets, particularly during crisis periods. As such, we do not perform separate estimations for the pre- and postcrisis subperiods, as the model accounts for these variations inherently. This approach aligns with the justification of Antonakakis et al. (2020) and Primiceri (2005).

Time-varying ımpacts of cryptocurrency, commodity, bond, and stock markets

We employ the time-varying impulse response function (IRF) to examine how the cryptocurrency market impacts the commodity, bond, and stock markets during various financial and geopolitical crises, such as the COVID-19 outbreak, the Russia–Ukraine war, and the Silicon Valley Bank collapse. This model allows us to observe how the impulse responses evolve when a positive shock is introduced at different points in time. We analyze the responses one day after the shock, from days 5–10 after, and beyond 10 days after the shock to capture short-term, medium-term, and long-term effects, respectively. The horizontal axis represents each time point, whereas the vertical axis shows the magnitude of the response of each variable to the shock. We can assess how quickly a shock propagates through financial markets by examining the reactions across these different periods.

Table 8 provides the parameter estimates for the TVP-VAR-SV model. The results show that all estimated posterior means fall within the 95% confidence interval. Additionally, the Geweke values and inefficiency factors confirm the effectiveness and

Table 8 TVP-VAR-SV parameter estimation results

Parameter	Mean	Stdev	95%L	95%U	Geweke	Inef
Panel A. Returns						
sb1	0.0227	0.0026	0.0183	0.0283	0.617	5.87
sb2	0.0226	0.0026	0.0182	0.0283	0.563	3.57
sa1	0.0849	0.0342	0.0428	0.1693	0.636	34.16
sh1	0.2454	0.104	0.0929	0.4954	0.006	53.62
sh2	0.2021	0.0944	0.0749	0.4315	0	116.6

robustness of the sampling results. Specifically, all Geweke values are below the critical threshold of 1.96, supporting the null hypothesis that the model converges in distribution to the posterior distribution. Furthermore, the low values of the inefficiency factors indicate that the number of iterations was sufficient for reliable posterior inference.

Figure 7 illustrates the impulse response of the commodity, bond, and stock markets to shocks originating from the cryptocurrency market during three key periods: the 2020 COVID-19 pandemic, the 2022 Russia–Ukraine conflict, and the 2023 Silicon Valley Bank collapse. Figure 8 illustrates time point-based impulse response of cryptocurrency, commodity, bond, and stock markets to a unit shock in returns and volatilities. As shown in Fig. 7, the impulse response of return and volatility spillovers from these markets to the cryptocurrency market rapidly increases, reaching its peak during periods of financial and geopolitical crisis.

Conclusion

This paper has thoroughly examined the intricate relationship between cryptocurrency adoption and traditional financial markets through a comprehensive analysis encompassing price and volume connectedness, descriptive statistics, nonlinearity tests, and the application of the TVP-VAR model with stochastic volatility, as proposed by Antonakakis et al. (2020) and based on the methodologies of Diebold and Yilmaz (2012, 2014). The empirical results reveal that the level of interconnectedness fluctuates between 20 and 70% for returns and between 40 and 80% for volatilities, with peaks observed during periods of significant market stress, such as in early 2018, 2020, and 2022. These peaks correspond to pivotal events, including cryptocurrency bubbles, the COVID-19 outbreak, the Russia–Ukraine conflict, and the Silicon Valley Bank crisis.

In particular, the interconnectedness index exceeded 80% in 2018, driven by the bursting of the cryptocurrency bubble, which was exacerbated by the spread of false

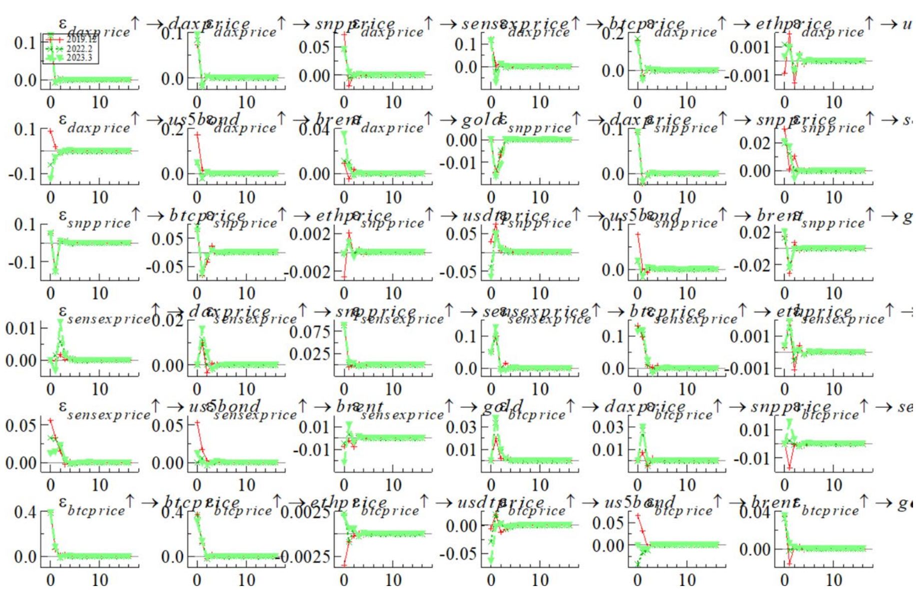


Fig. 7 Time-varying ımpulse response of cryptocurrency, commodity, bond, and stock markets to a unit shock in returns. *Note*: The red, black, and green lines represent the impulse responses at 1, 5, and 10 periods after a unit shock, respectively

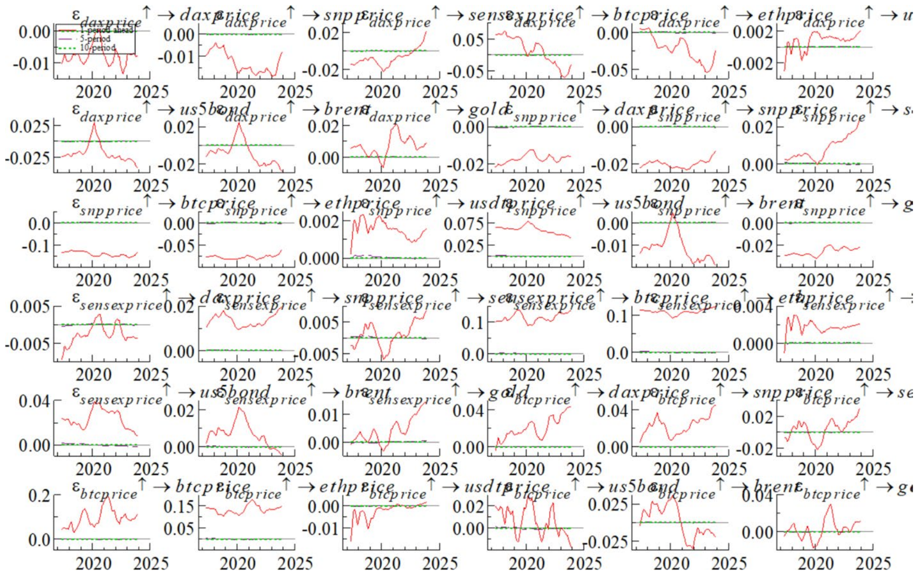


Fig. 8 Time point-based impulse response of cryptocurrency, commodity, bond, and stock markets to a unit shock in returns and volatilities. *Note*: The red, dark green, and bright green lines represent the impulse responses during the periods of the COVID-19 outbreak, the Russia–Ukraine conflict, and the Silicon Valley Bank crisis, respectively

news. Similarly, during the 2020 COVID-19 outbreak, the degree of interconnectedness peaked at over 65% because of the pandemic's profound impact on both major financial and cryptocurrency markets. These findings underscore the tendency for interconnectedness to intensify during periods of financial and geopolitical crises, reflecting the markets' collective responses to these stressors. As stated in Xu et al. (2024), investors often prioritize short-term financial stability over longer-term sustainable investments in times of high uncertainty. Consequently, this shift in focus prompts adjustments in investment strategies, leading to a marked and temporary rise in market connectedness. Overall, the total connectedness among international sustainable investments has increased notably during these extreme events, underscoring that the shocks from such occurrences profoundly impact the interrelations of various markets on a global scale.

Our analysis also identifies the SENSEX index, USDT, crude oil, and the 5-year U.S. bond market as primarily shock receivers within the system, whereas the DAX index, S&P 500 index, Bitcoin, and Ethereum generally serve as shock transmitters. Notably, gold, typically a shock receiver, transitions into a shock transmitter during periods of extreme turmoil, such as the COVID-19 outbreak and the Russia–Ukraine conflict, and in response to high inflationary pressures. Cryptocurrencies, particularly Bitcoin and Ethereum, emerge as significant shock transmitters during extreme events, suggesting that investors increasingly perceive these digital assets as safe havens alongside traditional assets such as gold.

In terms of volatility, our findings reveal that Bitcoin and Ethereum shifted from being volatility receivers to transmitters after mid-2021, a period marked by the ongoing impacts of the COVID-19 pandemic and the escalation of the Russia–Ukraine war. Additionally, stock markets are shown to contribute to cryptocurrency market volatility, whereas the cryptocurrency market, in turn, transmits volatility to the 5-year U.S. bond and commodity markets, especially during periods of heightened uncertainty.

Cryptocurrencies are also found to exhibit greater hedge effectiveness than gold and crude oil in portfolio management strategies. From the perspective of volatility portfolios, cryptocurrencies demonstrate greater efficiency, reinforcing their role as a potentially valuable component in diversified investment portfolios.

When our findings are compared with those of the literature, it becomes evident that financial and geopolitical risks significantly influence the dynamic connectedness between cryptocurrencies and traditional asset classes, with notable increases in volatility spillovers during crises. These results are consistent with the conclusions of previous studies by Bouri et al. (2020), Aslanidis et al. (2020), Mensi et al. (2021), Bakry et al. (2021), Nguyen (2022), and Zhao and Zhang (2023), all of which emphasize the heightened interconnectedness and volatility within the cryptocurrency market and its interplay with traditional financial assets during turbulent times.

In summary, our study corroborates the growing body of literature that documents increasing interconnectedness and volatility spillovers involving cryptocurrencies, particularly during periods of market stress. These insights highlight the necessity of closely monitoring the dynamic relationships between cryptocurrencies and other financial assets, especially during times of uncertainty and turmoil. The results of our study carry significant implications for market participants, particularly during crises when the benefits of diversification may shift. Investors need to carefully select diversification strategies that align with their risk preferences. The presence of significant skewness and kurtosis spillovers underscores the need to move beyond conventional risk measures focused solely on volatility. Asymmetric and tail risk dynamics, particularly during crisis episodes, reveal that cryptocurrencies not only transmit volatility but also contribute to systemic asymmetries. These insights are critical for both investors and regulators seeking to understand and mitigate nonlinear risks in interconnected financial systems. The insights provided by our research offer valuable guidance for cryptocurrency investors, enabling them to make informed decisions in portfolio management.

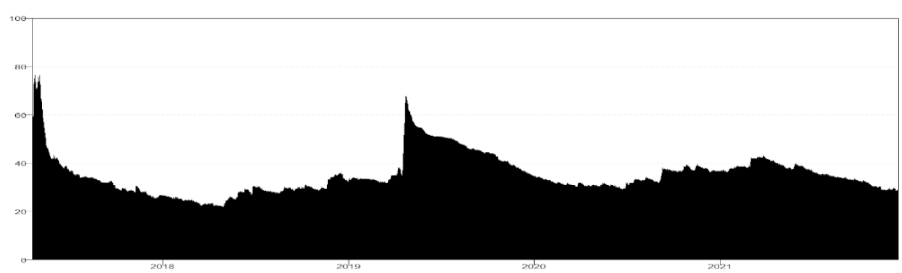
Cryptocurrency spillovers refer to the transmission of volatility or risk from crypto markets to traditional financial sectors or assets, such as banks, traditional markets, or stablecoins. As crypto assets integrate into the broader financial system, regulators are developing tools to manage systemic risk and reduce contagion. Regulators may require banks to hold additional capital buffers to absorb potential losses resulting from crypto volatility. The Basel Committee on Banking Supervision (BCBS) has established a framework for managing cryptoasset exposures, classifying them into two groups on the basis of their risk profiles and characteristics. Group 1 includes lower-risk cryptoassets, such as tokens and stablecoins, which adhere to standard risk-based capital requirements, along with a 2.5% infrastructure risk add-on. Group 2 consists of higher-risk assets such as Bitcoin, requiring banks to hold 1:1 capital backing and subject to a 1,250% risk weight. Banks can allocate only 1% of their Tier 1 capital to Group 2 assets, capped at 2%, with any excess deducted from their Common Equity Tier 1 (CET1) capital (BCBS 2023).

Moreover, our research underscores the importance of policymakers having appropriate regulatory tools to effectively oversee the cryptocurrency market and mitigate potential risks. For risk-seeking investors, cryptocurrencies may serve as potential diversifiers, whereas risk-averse investors should exercise caution and limit their exposure to cryptocurrencies to avoid excessive risk in extreme market conditions. Central Bank digital currencies (CBDCs) are increasingly recognized as vital tools for managing the systemic risks linked to cryptocurrencies in the global financial system. These risks include financial instability and regulatory challenges. As government-backed currencies, CBDCs offer a stable alternative to the volatility of cryptocurrencies, allowing central banks to maintain control over monetary policy. While cryptocurrencies operate independently and pose risks to savings and stability, CBDCs provide secure assets for investors, potentially influencing cryptocurrency market behavior (Auer et al. 2021; Le 2025). By ensuring a regulated alternative, CBDCs help maintain public confidence in the financial system and address the dangers of unregulated digital assets. Supervised by central banks, these digital currencies aim to create a reliable and secure alternative to the inherent volatility and decentralization found in cryptocurrencies. Their emergence as a government-backed option can increase stability and mitigate risks during market downturns, which is crucial for fostering trust in the financial system (Akin et al. 2023; Rizwan et al. 2025).

Our findings suggest that cryptocurrencies can act as diversifiers, with portfolios containing cryptocurrencies generally outperforming those without cryptocurrencies in portfolio optimization frameworks. By shedding light on the complex relationship between cryptocurrency and traditional financial markets, our study provides valuable information for both investors and policymakers, contributing to a deeper understanding of the evolving financial landscape.

Appendix

See Figs. 9, 10, 11, 12, 13, 14, 15, 16 and 17.



Panel B Dynamic total connectedness (Minnesota Priors)

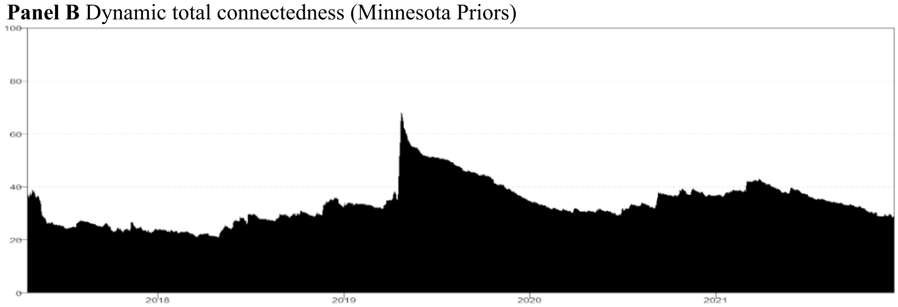


Fig. 9 Prior sensitivity analysis: dynamics of total connectedness index

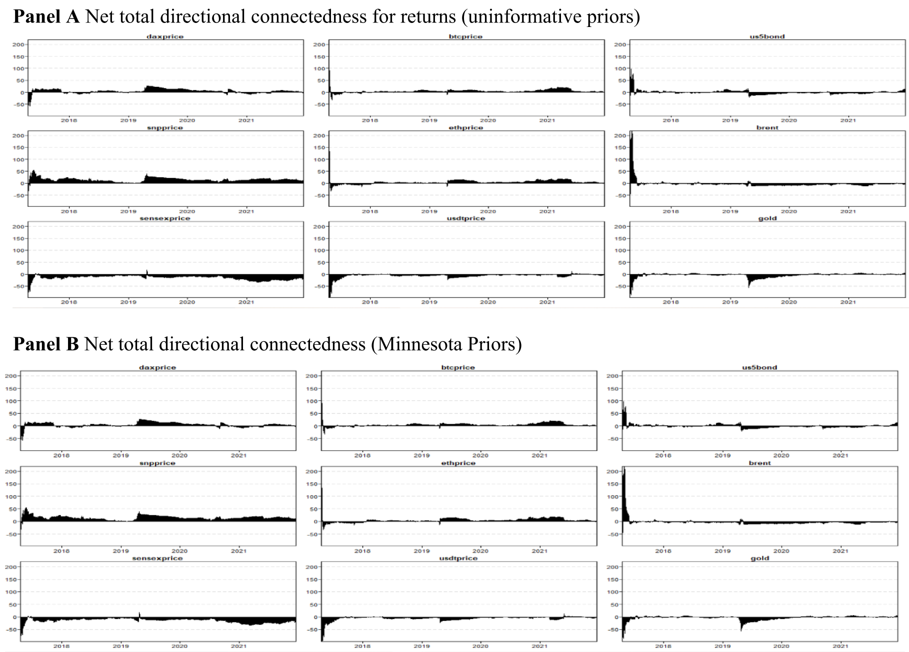


Fig. 10 Prior sensitivity analysis: dynamics of the net total directional connectedness

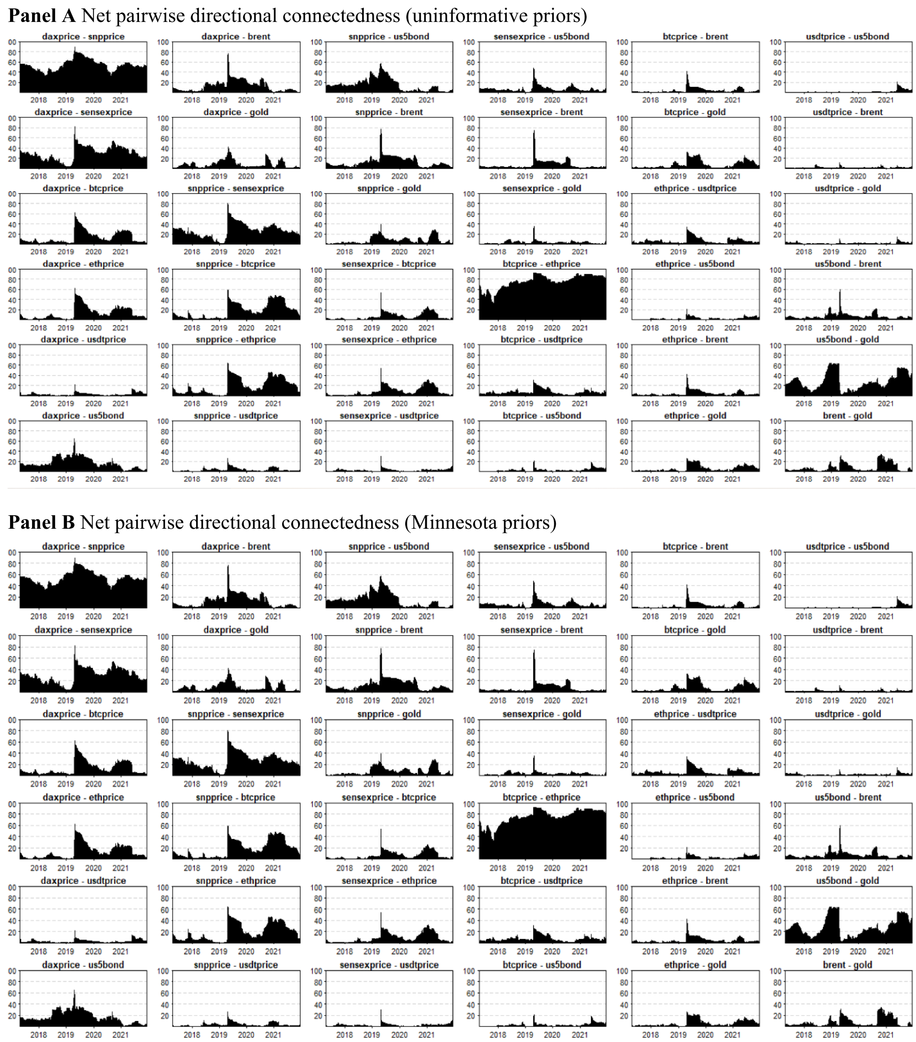


Fig. 11 Prior sensitivity analysis: net pairwise directional connectedness

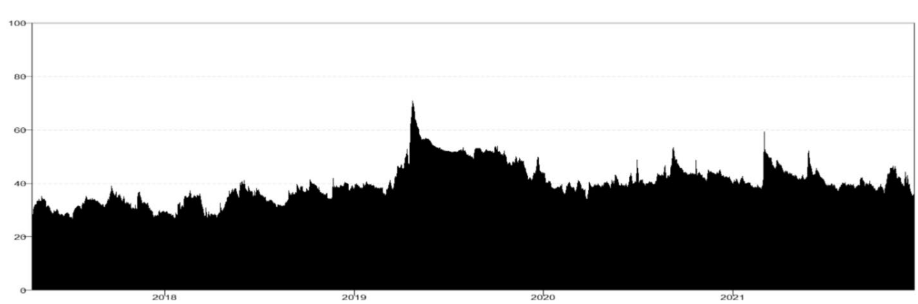


Fig. 12 Forgetting factor sensitivity analysis: total dynamic connectedness uncertainty

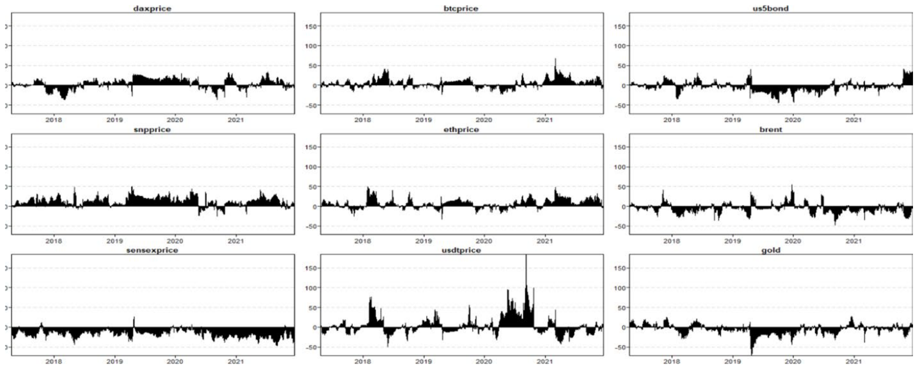


Fig. 13 Forgetting factor sensitivity analysis: net total directional connectedness uncertainty

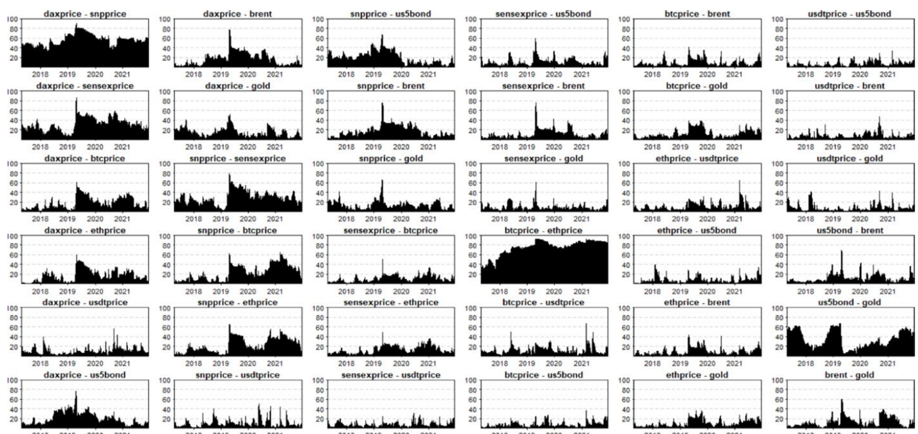


Fig. 14 Forgetting factor sensitivity analysis: net pairwise directional connectedness uncertainty

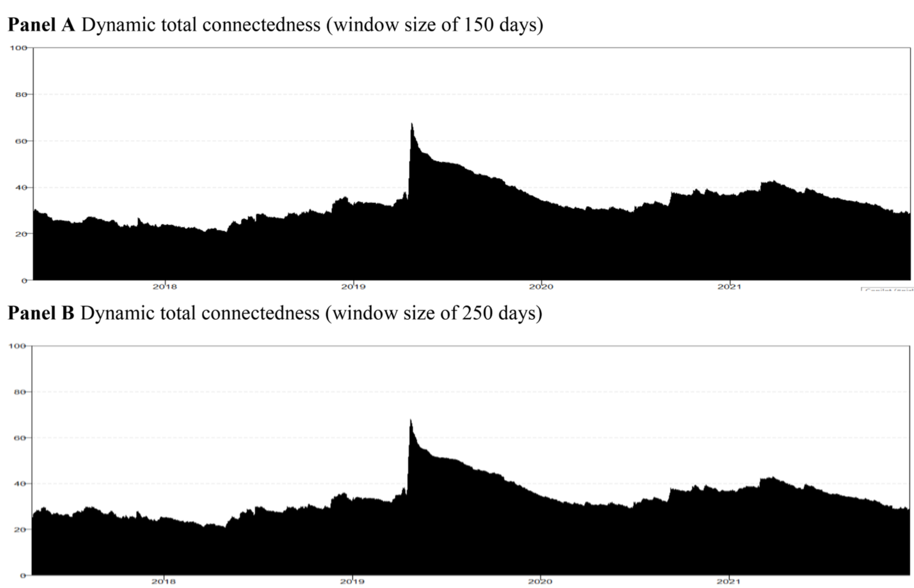


Fig. 15 Dynamics of total connectedness index. *Notes*: Results are based on a TVP-VAR model with lag length of order one (BIC) and a 20-step-ahead forecast and different rolling window

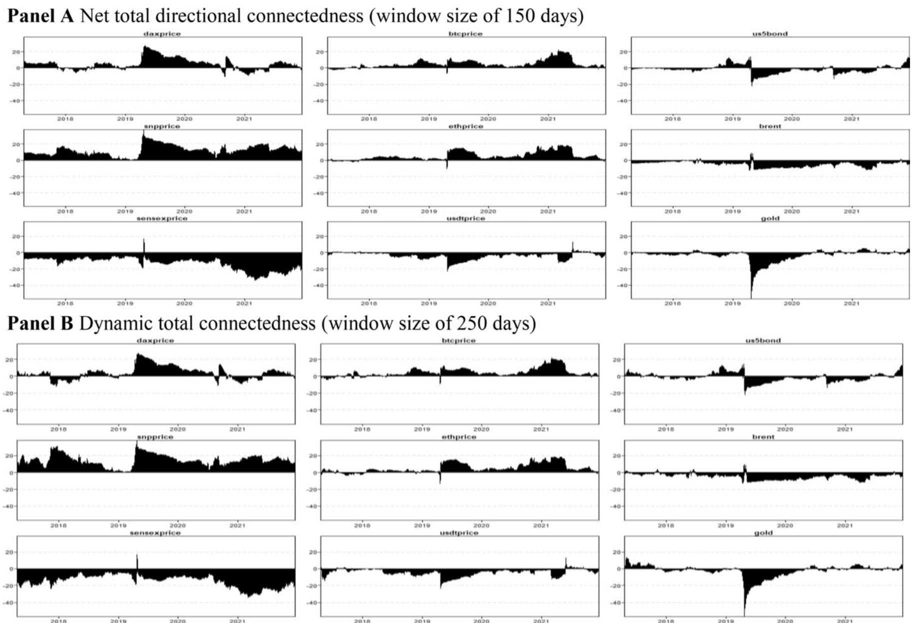


Fig. 16 Dynamics of the net total directional connectedness. İndex. *Notes*: The results are based on a TVP-VAR model with a lag length of order one (BIC) and 20-step-ahead forecasts and different rolling windows

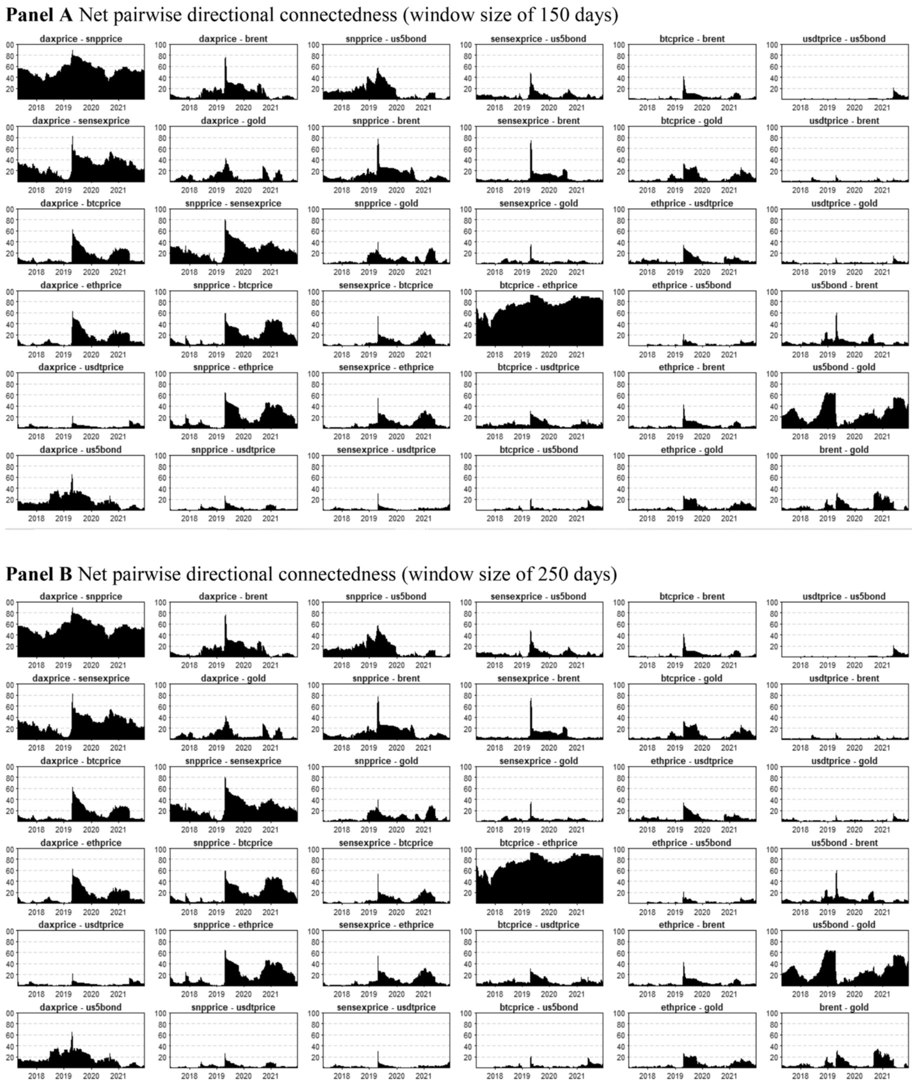


Fig. 17 Net pairwise directional connectedness. *Notes*: Results are based on a TVP-VAR model with a lag length of order one (BIC) and a 20-step-ahead forecast and different rolling windows

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Competing interests
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